

Missouri Field Guide

Edible Plants · Fungi · Wildlife · Fish · Animal Tracking · Trees · Danger Zone

A comprehensive field reference for Missouri's outdoors —
identification, ecology, edibility, utility, and safety

2024 Edition

Wildflowers & Plants

31 entries

Wildflowers & Plants American Persimmon



The American Persimmon (*Diospyros virginiana*) is a tough, adaptable native tree found across all of Missouri in dry upland woods, old fields, roadsides, and forest edges. Trees reach 20–60 feet with deeply furrowed, blocky bark resembling alligator hide — one of the most distinctive bark patterns in Missouri. Small, creamy-white flowers bloom in May–June; orange-yellow to purple fruit (3/4–1.5 inches) ripens in October after first frost.

■ Utility & Value

Unripe persimmons contain high tannin levels causing a severe astringent pucker; ripe ones — especially after hard frost — are sweet, rich, and date-like. Eat fresh, make persimmon pudding (a Missouri Ozark tradition), dry into fruit leather, or ferment into beer and wine. The wood is extremely hard and dense — historically used for golf club heads, billiard cues, and weaving shuttles. White-tailed deer, raccoons, opossums, foxes, and turkeys all feed heavily on fallen persimmons in autumn, making persimmon trees excellent hunting stand locations.

Key Facts

- Do NOT eat until fully ripe and soft — tannins cause severe mouth puckering
- Frost accelerates ripening but isn't strictly required — wait for soft, wrinkled fruit
- Trees are dioecious — you need male and female trees nearby for fruit production
- Persimmon pudding is a traditional Ozark dish still made at festivals today
- Deer, turkey, and opossum sign near persimmon trees indicates prime hunting locations
- Inner seed coat cross-section predicts winter severity in folklore (spoon = snow to shovel)

Wildflowers & Plants Black Walnut



Black Walnut (*Juglans nigra*) is one of Missouri's most valuable native trees, reaching 70–100+ feet in rich bottomland and upland soils statewide. The compound leaves (15–23 leaflets) emit a distinctive spicy-citrus odor when crushed. Fruits are round, 1.5–2.5 inch green husks that turn black when ripe in September–October. The husks stain everything they touch an indelible dark brown.

■ Utility & Value

The nutmeat is intensely flavored — bolder and earthier than commercial English walnut — and prized in baked goods, ice cream, and candy. To harvest: let husks blacken and soften (or drive over them), remove husks wearing gloves, wash shells, cure for 2–3 weeks, then crack with a hammer. The tree produces juglone, a chemical that inhibits many plants growing nearby. Black walnut lumber is among North America's most valuable hardwoods — prized for gunstocks, furniture, and veneer. Husk tea was used as a natural dye and antifungal treatment.

Key Facts

- Husks stain skin brown for weeks — wear gloves when processing
- Juglone toxin kills tomatoes, apples, and many garden plants within 50–80 feet of roots
- Black walnut lumber: a single mature tree can be worth thousands of dollars
- Cure nuts 2–3 weeks after husking; store in shell up to a year in cool dry place
- Husk extract is a traditional remedy for ringworm and other fungal infections
- Squirrels bury more nuts than they find — walnut trees depend on squirrels for dispersal



Common Cattail (*Typha latifolia*) and Narrowleaf Cattail (*T. angustifolia*) grow in marshes, pond margins, ditches, stream banks, and wet disturbed areas throughout Missouri. The flat, ribbon-like leaves reach 3–8 feet and the iconic brown cylindrical seed heads (the 'cat tail') are 4–8 inches long. Nearly every wetland margin in Missouri supports dense cattail stands. Often dismissed as a nuisance plant, cattail is actually one of the most useful plants in North America.

■ Utility & Value

Cattails offer food year-round at different growth stages: in spring, peel the young green shoots for a crisp, mild vegetable eaten raw or cooked like asparagus. In late spring, green pollen spikes (male flowers above the brown female head) can be boiled and eaten like corn. In early summer, harvest yellow pollen by shaking into a bag — use as flour supplement high in protein. Fall and winter: dig rhizomes, wash, and pound in water to release starchy flour — highly nutritious. Dried leaves weave excellent baskets, mats, and chair seats. Fluffy seed heads make excellent tinder and insulation.

Key Facts

- Four distinct edible stages: shoots (spring), green spikes (late spring), pollen (early summer), rhizome starch (fall-winter)
- Pollen is high in protein — mix into pancake batter as flour extender
- Fluffy seed down was used as diaper material and fire tinder by Indigenous peoples
- Leaves weave baskets, mats, and emergency shelter roofing
- Cattail marshes provide critical nesting habitat for red-winged blackbirds and muskrats
- Missouri muskrats eat cattail rhizomes — look for 'eat-outs' (open water circles in marshes)



The Common Dandelion (*Taraxacum officinale*) is perhaps the most underrated edible plant in Missouri. Found in lawns, fields, roadsides, and disturbed ground statewide, this European native has been used as food and medicine for thousands of years. The deeply toothed leaves form a basal rosette; bright yellow composite flowers bloom spring through fall. The hollow flower stalk exudes milky latex when broken. Seed heads form the familiar white puffballs.

■ Utility & Value

Every part of the dandelion is edible. Young spring leaves (before flowering) are the mildest — use raw in salads or sauté with garlic and olive oil. Older leaves become more bitter but are still nutritious — boil and change water once to reduce bitterness. Flowers can be battered and fried, added to salads, or used to make dandelion wine (a Missouri rural tradition). Roasted roots make a caffeine-free coffee substitute. Dandelion is extraordinarily nutritious: higher in vitamin A than carrots, more calcium than milk by weight, rich in vitamins C and K, iron, potassium, and inulin (a prebiotic fiber). Historically used as a diuretic and liver tonic.

Key Facts

- More nutritious per ounce than most cultivated vegetables — higher Vitamin A than carrots
- Roots roasted at 350°F for 40 min, then ground, make a coffee-like caffeine-free drink
- Dandelion wine: ferment flowers with sugar, yeast, and citrus — a traditional Ozark recipe
- Young spring leaves are mildest; harvest before flowers open for salads
- Inulin in roots feeds beneficial gut bacteria — a natural prebiotic
- One of the first spring flowers — critical early nectar source for bees and butterflies

Death Camas



Death Camas (*Anticlea elegans*, formerly *Zigadenus elegans*) and related species (*A. nuttallii*) grow in prairies, open woodlands, and glades in Missouri, particularly in the Ozarks and northern prairies. The grass-like, V-shaped basal leaves and dense raceme of creamy-white flowers can resemble wild garlic, wild onion, or edible camas — a potentially fatal confusion. The crucial difference: Death Camas has absolutely no onion or garlic odor.

■ Utility & Value

All parts — especially the bulb — contain toxic steroidal alkaloids (zygacine and related compounds) that cause a sharp burning sensation in the mouth, followed by salivation, nausea, vomiting, slow heart rate, low blood pressure, muscle weakness, and potentially fatal cardiac or respiratory failure. Livestock losses from Death Camas are documented across the Great Plains. There is no edible use. The single most important rule: if it doesn't smell like onion or garlic, do not eat it. This rule has prevented countless deaths.

Key Facts

- LOOKS like wild onion/garlic but has ZERO onion odor — the critical distinction
- All parts toxic — bulb most concentrated; causes cardiac and respiratory failure
- Never eat any 'onion-like' bulb plant that lacks a true onion smell
- Responsible for significant livestock losses across the Midwest every spring
- Flowers are creamy white on a tall raceme — bloom May–July
- Grows in prairies and glades where wild onions also grow — overlapping habitat increases risk

Elderberry



American Elderberry (*Sambucus canadensis*) grows vigorously along streams, roadsides, forest margins, and disturbed ground throughout Missouri, often forming dense multi-stemmed shrub colonies 5–12 feet tall. Large compound leaves with 5–11 leaflets and flat-topped clusters (cymes) of tiny white flowers bloom June–July. Dark purple-black berries ripen in August–September in heavy drooping clusters. The pithy stems are soft and hollow.

■ Utility & Value

Elderberries are among the most medicinally documented wild plants in North America. The berries are rich in vitamin C, vitamin B6, and iron, and contain compounds (anthocyanins, sambunigrin) shown in clinical trials to reduce flu duration and severity. Make syrup, wine, jam, or pie — always cook before consuming as raw berries can cause nausea. Flowers are edible and delicious battered and fried as fritters or steeped into elderflower cordial. The hollow stems were used by Indigenous peoples as flutes, blowguns, and taps for maple syrup.

Key Facts

- Raw berries cause nausea — cook or dry before consuming
- Elderflower cordial: soak flower clusters in sugar-water for 24–48 hours
- Red elderberry (*S. racemosa*) is toxic — avoid; only dark purple/black berries are edible
- Stems, leaves, bark, and roots contain cyanogenic glycosides — do not eat
- Elderberry syrup: simmer 1 cup berries in 2 cups water, strain, add honey
- Flowers bloom June–July; harvest berries August–September when fully dark purple

Jimsonweed / Datura



Jimsonweed (*Datura stramonium*) is a coarse annual weed, 2–5 feet tall, found in disturbed ground, farmyards, roadsides, and waste areas throughout Missouri. The large, irregular, foul-smelling leaves and spectacular large white to pale violet trumpet-shaped flowers (3–4 inches) are distinctive. The fruit is a spiny green egg-shaped capsule. All parts contain powerful tropane alkaloids — atropine, scopolamine, and hyoscyamine.

■ Utility & Value

All parts are highly toxic with a narrow margin between a 'hallucinogenic' dose and a lethal one. *Datura* causes severe anticholinergic syndrome: racing heart, fever, dry mouth, dilated pupils, extreme confusion, violent delirium, and seizures. People do not 'enjoy' *Datura* experiences — survivors typically describe horrifying, indistinguishable-from-reality hallucinations lasting 12–48 hours with total amnesia afterward. Deaths occur regularly. Despite this, attempts to use it recreationally continue — often with fatal or permanently disabling results. There is no safe recreational dose.

Key Facts

- Contains atropine, scopolamine, and hyoscyamine — all extremely toxic alkaloids
- Causes violent, terrifying delirium, not pleasant hallucinations — survivors often describe it as a nightmare
- Margin between hallucinogenic and lethal dose is dangerously narrow
- Spiny seed capsules persist through winter — easy winter ID
- Foul smell of crushed leaves is a warning sign — repels most animals
- Atropine (purified from *Datura*) is a legitimate cardiac medication — but not from the plant directly

Lamb's Quarters



Lamb's Quarters (*Chenopodium album*) is an annual weed common in gardens, disturbed ground, fields, and roadsides throughout Missouri from spring through fall. Plants grow 1–5 feet tall with diamond- to goosefoot-shaped leaves that appear powdery or mealy on the undersides and young growing tips — this powdery coating is the key identification feature. Small greenish flowers in dense clusters appear midsummer.

■ Utility & Value

Lamb's quarters is arguably the most nutritious leafy vegetable you can forage in Missouri — it outperforms spinach in protein, calcium, iron, and vitamin content. Young plants and growing tips can be eaten raw in salads, but the mild flavor really shines when briefly sautéed with oil and garlic like spinach. Seeds (tiny black) ripen in fall and can be ground into flour or cooked as a grain — they're related to quinoa, which is in the same plant family. It wilts down dramatically when cooked, so harvest generously. This plant is so nutritious and abundant that it was a staple food for many Native American and early American cultures.

Key Facts

- More nutritious than spinach — higher in protein, calcium, iron, and vitamins A and C
- Powdery coating on leaves is the key ID feature — not harmful, just a surface secretion
- Closely related to quinoa (*Chenopodium quinoa*) — seeds ground into flour are equally nutritious
- Available from May through October — one of the longest-season edible weeds
- Oxalic acid present in large quantities — not a concern for typical serving sizes
- Often found in garden beds — a welcome 'weed' if you know what it is

Mayapple



Mayapple (*Podophyllum peltatum*) is one of Missouri's most distinctive forest floor plants, forming dense colonies of umbrella-like leaves 12–18 inches tall in rich, moist hardwood forests statewide. Single-stemmed plants have one leaf and no flower; two-stemmed plants have two leaves and produce a single waxy white flower in April–May hidden beneath the leaf canopy. The lemon-yellow, egg-shaped fruit (1–2 inches) ripens in late July–August.

■ Utility & Value

Only the fully ripe yellow fruit is edible, and only in moderation — the seeds inside are toxic. The ripe fruit has a complex, tropical flavor with hints of strawberry, lemon, and guava. Make jelly, add to fruit salads, or eat carefully fresh (spitting out seeds). All other parts — roots, leaves, stems, unripe fruit, and seeds — contain podophyllotoxin, a powerful cytotoxic compound. Purified podophyllotoxin is the basis for etoposide, an important chemotherapy drug used to treat testicular cancer and small-cell lung cancer. This makes Mayapple one of the most medically significant Missouri wildflowers.

Key Facts

- Only ripe yellow fruit is edible — all other parts including seeds are toxic
- Podophyllotoxin from roots is the basis for etoposide, a critical chemotherapy drug
- Colonial growth — one patch may be a single genetic individual spreading via rhizomes
- Fruit ripens late July–August; pick when soft, yellow, and fragrant
- Seeds inside ripe fruit are toxic — eat flesh only, spit seeds
- Native peoples used root preparations carefully as a purgative and to treat warts

Moonseed



Canada Moonseed (*Menispermum canadense*) is a native climbing vine found in rich, moist woodlands and stream banks throughout Missouri. The large, rounded to shallowly lobed leaves superficially resemble grape leaves. In late summer, it produces drooping clusters of dark blue-black berries strikingly similar in appearance to wild grapes. This resemblance is the primary danger — it grows intertwined with actual wild grapes.

■ Utility & Value

All parts of Moonseed are toxic, containing alkaloids including dauricine that can cause respiratory paralysis. The berries contain a distinctive crescent-shaped (moon-shaped) seed — the source of the common name and the single most reliable identification feature distinguishing it from wild grape. Wild grape seeds are round; Moonseed seeds are crescent-shaped. Grape leaves also have tendrils; Moonseed does not. Children are particularly at risk from these attractive berries. Always cut open and examine the seed before eating any wild grape-like berry.

Key Facts

- Deadly lookalike for wild grapes — grows intertwined with true grape vines
- KEY DISTINCTION: Moonseed has crescent-shaped seed; grape has round seed
- Moonseed has NO tendrils; grapes have tendrils at leaf nodes
- All parts toxic — alkaloids can cause respiratory paralysis
- Children attracted to grape-like berries — educate them before letting them forage independently
- Cut open every berry before eating any wild grape to verify round seed



Bittersweet Nightshade (*Solanum dulcamara*) and Black Nightshade (*S. nigrum/americanum*) are two toxic weedy plants common in Missouri's disturbed areas, woodland edges, and gardens. Bittersweet Nightshade is a vine with purple flowers and red oval berries; Black Nightshade is a low annual with white flowers and small black berries. Both are related to tomatoes and potatoes — and equally toxic in their berries and unripe fruit.

■ Utility & Value

All Nightshade berries in Missouri should be considered toxic unless positively identified by an expert. Solanine, the primary toxin, causes nausea, vomiting, diarrhea, confusion, and in large doses, seizures and coma. Unripe (green) berries of all species are most toxic. The ripe black berries of Black Nightshade are cooked in some cultures (*S. nigrum* pie is traditional in South Africa), but this is not recommended without expert identification, as several toxic species are similar. Avoid all small red, black, or white berries on nightshade-type plants.

Key Facts

- Purple flowers with yellow stamens forming a beak = classic *Solanum* flower pattern
- Bittersweet Nightshade berries ripen from green to yellow to red — all stages on one plant
- Related to tomatoes, potatoes, eggplant — all in family Solanaceae
- Unripe green berries most toxic — solanine causes GI distress and neurological effects
- Common in backyards and gardens statewide — educate children to avoid red berries
- Name 'bittersweet' refers to the taste: initially bitter, then sweet — do not taste-test



Ohio Buckeye (*Aesculus glabra*) is Missouri's official state tree and a medium-sized deciduous tree reaching 20–40 feet, found in rich bottomlands, moist upland slopes, and stream valleys primarily in central and western Missouri. The distinctive opposite, palmately compound leaves (5–7 leaflets) and spiny husked nuts containing 1–3 smooth brown seeds (buckeyes) are unmistakable. Creamy-yellow flowers in spring are followed by leathery spiny capsules in fall.

■ Utility & Value

Ohio Buckeye seeds contain aesculin and saponins — toxic compounds causing nausea, vomiting, muscle weakness, and paralysis. They are not edible raw. Native peoples processed them by grinding, leaching with water for extended periods, and roasting — a laborious process that neutralized toxins but was used only in times of scarcity. The smooth, mahogany-brown 'lucky buckeye' seeds are carried as good luck charms throughout Missouri and Ohio — a deep Midwestern cultural tradition. The wood is soft and light, used for artificial limbs historically. Flowers are important for early spring pollinators.

Key Facts

- Missouri's official State Tree
- Seeds (buckeyes) are toxic — do not eat raw; historically processed as famine food only
- Carrying a buckeye in your pocket is a Missouri/Ohio good-luck tradition
- One of the earliest trees to leaf out in spring and first to drop leaves in fall
- Compound leaves with 5–7 leaflets distinguish it from all other Missouri trees
- Flowers attract early-season hummingbirds and bumblebees



The Pawpaw (*Asimina triloba*) is Missouri's largest native fruit and one of North America's best-kept edible secrets. It grows as an understory tree or large shrub, typically 15–30 feet tall, forming dense clonal colonies (patches) along stream banks, rich bottomland woods, and shaded ravines statewide. The large, drooping leaves — up to 12 inches long — give a tropical appearance strikingly out of place in the Midwest. Dark maroon, bell-shaped flowers appear in April before the leaves, and the fruit ripens August through October.

■ Utility & Value

Ripe pawpaws have custard-like, yellow-orange flesh tasting of banana, mango, and vanilla — rich in calories, vitamin C, magnesium, iron, and antioxidants. They can be eaten raw, blended into smoothies, baked into bread or ice cream, or fermented. Indigenous peoples including the Osage and Shawnee depended on them heavily. The bark and seeds contain acetogenins, compounds under study for anti-cancer properties. Fruit doesn't ship well, so wild-harvested pawpaws aren't commercially available — finding them in the wild is genuinely rare and valuable.

Key Facts

- Fruit ripens Aug–Oct; squeeze-test for softness like a ripe avocado
- Leaves and twigs emit a strong green-pepper odor when crushed
- Zebra swallowtail butterfly larvae feed exclusively on pawpaw leaves
- Clonal patches can persist for centuries in the same spot
- Seeds are toxic — do not eat; bark tea was used historically as an insecticide
- Missouri populations concentrated in Ozark stream valleys and Mississippi bottomlands



Poison Hemlock (*Conium maculatum*) is a non-native biennial weed common in disturbed areas, roadsides, fields, and waste places throughout Missouri. It grows 3–8 feet tall with hollow, smooth stems covered in distinctive purple-red blotches, ferny compound leaves with a disagreeable musty odor when crushed, and flat-topped white flower clusters. This is the plant used to execute Socrates in 399 BC.

■ Utility & Value

All parts are highly toxic — contains coniine and related alkaloids that cause ascending paralysis. Unlike Water Hemlock (which kills by convulsion), Poison Hemlock kills by respiratory paralysis as muscles fail from the feet upward. Death can occur within hours of ingestion. It has been mistaken for wild carrot, parsley, and Queen Anne's Lace. The musty, unpleasant odor when leaves are crushed is an important warning. Contact with sap can cause dermatitis. Children have been poisoned using the hollow stems as blowguns — never do this.

Key Facts

- Killed Socrates — one of history's most historically documented poisonings
- Purple-blotched hollow stem + musty odor when crushed = key identification
- Causes ascending paralysis — respiratory failure is the cause of death
- Children have died using hollow stems as blowguns — never handle casually
- Common along Missouri roadsides and disturbed areas — very widespread
- Confused with Wild Carrot (Queen Anne's Lace) — always check for purple stem blotches

Poison Ivy



Poison Ivy (*Toxicodendron radicans*) is arguably Missouri's most important plant to recognize — it grows in every county in every habitat type statewide as a ground vine, shrub, or climbing vine that can ascend 50+ feet up tree trunks using dense aerial rootlets that give the trunk a 'hairy rope' appearance. Leaves are always in groups of three, leaflets pointed with irregular teeth or lobes. Leaves are shiny and reddish when young, green and variable in summer, and brilliant red-orange in fall. White berries persist through winter.

■ Utility & Value

There is no edible use. About 85% of people are allergic to urushiol, the oily resin present in all parts of the plant year-round — including dead stems in winter. Contact causes intensely itchy blistering rash in 12–72 hours. Smoke from burning poison ivy can cause severe lung irritation and systemic reaction — never burn it. Wash with soap and cold water immediately after suspected contact. Calamine lotion, hydrocortisone cream, and oral antihistamines manage symptoms. Severe reactions (face swelling, difficulty breathing) require emergency medical care. Birds eat the berries without harm.

Key Facts

- 'Leaves of three, let it be' — the universal Missouri warning
- Urushiol remains active on dead plants, tools, and clothing for years
- NEVER burn Poison Ivy — smoke inhalation can cause life-threatening lung inflammation
- Wash exposed skin with cold water and soap within 10 minutes to reduce reaction
- White berries are eaten by 60+ bird species — birds are immune to urushiol
- Allergic sensitivity increases with repeated exposures — don't assume you're immune

Poison Sumac



Poison Sumac (*Toxicodendron vernix*) is a shrub or small tree 5–20 feet tall confined to swampy, boggy, and wet woodland habitats in Missouri — primarily the Bootheel and Mississippi lowlands. The pinnately compound leaves (7–13 leaflets with smooth margins) are glossy and smooth. The critical distinguishing feature from edible sumacs: Poison Sumac has WHITE to pale gray-green berries hanging in loose drooping clusters, while all edible sumacs have RED upright berry clusters.

■ Utility & Value

Poison Sumac contains urushiol — the same compound as Poison Ivy, but often considered more potent because it grows in wetlands where people may grab it unexpectedly. Contact causes the same intensely itchy, blistering rash as Poison Ivy. Like Poison Ivy, all parts are toxic year-round including dead stems and smoke from burning. It is less commonly encountered than Poison Ivy due to its specific wetland habitat, but the reaction can be more severe. Know the white-berry / wet-habitat combination as the definitive warning signs.

Key Facts

- WHITE hanging berry clusters in WETLANDS = Poison Sumac — always
- RED upright berry clusters = safe Smooth or Staghorn Sumac
- Contains urushiol — same toxin as Poison Ivy, potentially more concentrated
- Confined to wet, boggy habitats in Missouri's Bootheel and Mississippi floodplain
- Burning releases toxic smoke — same danger as burning Poison Ivy
- Less common than Poison Ivy but more dangerous where encountered due to surprise factor



Common Pokeweed (*Phytolacca americana*) is a large, dramatic native plant growing 4–10 feet tall with thick magenta-pink stems, huge elliptical leaves up to 12 inches long, and long dangling clusters of dark purple-black berries. It grows in disturbed ground, fields, woodland edges, and fence rows throughout Missouri. Despite being toxic, it is one of the most culturally significant plants in Missouri and Appalachian folklore.

■ Utility & Value

Pokeweed is toxic — roots most severely (mitogenic proteins causing potentially fatal blood cell changes), then berries, then stems and leaves. However, very young spring shoots under 6 inches can be rendered edible by boiling in two to three changes of water — this 'poke sallet' (salad) was a traditional Ozark survival food eaten by generations. Modern foragers generally advise against it given the risk. The berry juice was used as ink and dye (pokeberry ink was used by Confederate soldiers to write letters during the Civil War). The berries are eaten by mockingbirds, catbirds, and many other songbirds that are resistant to the toxins.

Key Facts

- Root is most toxic — never eat; causes severe vomiting, bloody diarrhea, and potentially death
- Young shoots under 6 inches can be eaten after boiling in 3 changes of water — traditional 'poke sallet'
- Berries are beautiful but toxic to humans; songbirds eat them without harm
- Berry juice was used as ink — 'pokeberry ink' letters survive from the Civil War
- Seeds can remain viable in soil for 40+ years — established plants are very persistent
- Pink stems and purple-black berry clusters make this one of Missouri's most striking plants



Common Purslane (*Portulaca oleracea*) is a low-growing, mat-forming annual succulent found in gardens, disturbed soil, sidewalk cracks, and cultivated fields throughout Missouri from June through September. The thick, fleshy stems are reddish; the paddle-shaped succulent leaves are smooth and shiny. Tiny yellow flowers open only in morning sunlight. Most gardeners pull it as a weed — a significant mistake.

■ Utility & Value

Purslane is the richest plant source of omega-3 fatty acids (ALA) available in the wild — higher than flaxseed per fresh weight and remarkable for a leafy green. It is also high in vitamins A, C, and E, magnesium, calcium, and antioxidants. The flavor is mildly tart and succulent — eat raw in salads where it provides pleasant crunch, sauté briefly like spinach, add to soups, or pickle in vinegar. In Mediterranean and Middle Eastern cuisine it's a prized vegetable sold in markets. The seeds can be ground into flour. Eaten globally for millennia — Gandhi counted purslane among his favorite foods.

Key Facts

- Highest plant source of omega-3 fatty acids — remarkable for a leafy green
- Succulent stems and leaves have a pleasant lemony-tart flavor
- Seeds viable in soil for 40+ years — removing before seeding saves future weeding
- Available June–September — one of summer's most reliable wild edibles
- Close relative of moss rose (*Portulaca grandiflora*) grown as a garden ornamental
- Widely eaten in Mexican, Greek, Turkish, and Chinese cuisines as a vegetable

Red Mulberry



Red Mulberry (*Morus rubra*) is a medium-sized native tree reaching 20–70 feet, found in rich bottomland woods, stream banks, and forest edges throughout Missouri. The leaves are variable — some unlobed, others with 2–3 lobes on the same branch — and sandpaper-rough on top. Inconspicuous flowers appear in April; elongated blackberry-like fruits (1–1.5 inches) ripen May–June, progressing from white to red to dark purple-black.

■ Utility & Value

Ripe mulberries are sweet and mild with a flavor reminiscent of blackberry and grape. They can be eaten fresh, made into pies, jam, wine, or dried. The tree produces prolifically — a single large tree can yield hundreds of pounds of fruit — but it ferments quickly once fallen, so harvest often. Berries stain everything purple. Deer, turkey, raccoon, opossum, fox, and virtually every fruit-eating bird gorge on mulberries in June, making trees excellent wildlife observation spots. Leaves were the historical food of silkworms in early American silk experiments.

Key Facts

- Ripens May–June — earlier than most wild fruit; spread a tarp under tree to collect
- Stains skin and clothing dark purple — wear old clothes when harvesting
- Wildlife magnet: deer, turkey, raccoon, fox, and 40+ bird species visit fruiting trees
- White Mulberry (*M. alba*, invasive) hybridizes with native Red Mulberry — threatening it
- Missouri's largest Red Mulberry trees exceed 3 feet in diameter in old-growth bottoms
- Unripe white/green fruits contain hallucinogenic compounds — only eat fully ripe dark fruit

Serviceberry / Juneberry



Serviceberry (*Amelanchier arborea* and *A. sanguinea*) grows as a large shrub or small tree 10–25 feet tall at forest edges, stream banks, and dry upland woods throughout Missouri. One of the earliest flowering trees in spring — its clusters of delicate white flowers appear in March–April before the leaves, often while snow is still possible. Small blueberry-sized fruits ripen in June (hence 'Juneberry'), turning from red to dark purple. In the Ozarks, it often grows on rocky slopes and cedar glades.

■ Utility & Value

Serviceberries are among the finest-flavored wild fruits in Missouri — sweet with an almond-cherry undertone from the seeds. Use exactly like blueberries: eat fresh, make pie, jam, muffins, or wine. Nutritionally similar to blueberries — high in anthocyanins, iron, manganese, and fiber. Native peoples dried them like raisins for winter storage and combined them with dried meat and fat to make pemmican — one of North America's original high-energy trail foods. The blooming of serviceberry was historically the signal for spring fishing and planting to begin.

Key Facts

- Earliest fruit to ripen in Missouri — ready in June before almost any other wild fruit
- Seeds have a pleasant almond flavor — don't pit them when making jam
- Traditional pemmican ingredient: dried serviceberries mixed with rendered fat and jerky
- Competition is fierce — birds strip trees within days of ripening; check daily
- Early white flowers are an important nectar source when few other plants bloom
- Name 'Serviceberry' comes from colonial times when blooming meant ground was soft enough for burial services

Shagbark Hickory Nuts



Shagbark Hickory (*Carya ovata*) is immediately recognizable by its dramatic shaggy bark — long curved gray plates that peel away from the trunk in distinctive curls, earning it the most descriptive name in Missouri's forest. It grows 60–80 feet in upland and dry woods throughout the state. The compound leaves have 5 leaflets (the terminal 3 are largest). Round to oval nuts inside thick husks that split into 4 sections when ripe in September–October.

■ Utility & Value

Shagbark hickory nuts have the highest caloric density of any North American nut — rich in fat, protein, and minerals. The flavor is sweet and rich, similar to pecan (a close relative). Crack shells, extract meat, and use in pies, cookies, and candy. Indigenous peoples boiled cracked nuts in water to extract 'hickory milk' — a nutrient-dense fat used in cooking. Hickory wood burns extremely hot and long — the gold standard for smoking meats and the historical choice for tool handles, wagon wheels, and baseball bats. Bark tea was used for fever and rheumatism.

Key Facts

- Most calorie-dense native nut in North America — critical autumn food for deer, turkey, squirrel
- Hickory smoke is the standard for barbecue — chips sold commercially for \$8+/lb
- Shaggy bark is habitat for many bat species that roost beneath the plates
- Nuts ripen Sept–Oct; fallen husks turn black and split open at 4 seams
- Wood density makes it ideal for handles — hickory axe handles outlast all alternatives
- Pecan (*C. illinoensis*) is also native to Missouri's southeastern river bottoms

Smooth Sumac



Smooth Sumac (*Rhus glabra*) and Staghorn Sumac (*R. typhina*) grow as multi-stemmed shrubs or small trees 5–15 feet tall at roadsides, old fields, and forest edges throughout Missouri. The compound leaves (11–31 leaflets) turn brilliant scarlet in autumn — among Missouri's finest fall color. Dense, upright clusters of fuzzy red berries (drupes) ripen in late summer and persist through winter, providing critical food for wildlife. **IMPORTANT:** Do not confuse with Poison Sumac (white berries, wetland habitat, glossy leaves).

■ Utility & Value

The red berries of Smooth Sumac are the basis for 'sumac-ade' — a traditional drink made by soaking berry clusters in cold water for 20–30 minutes, then straining. The result is a pink, lemonade-like beverage high in vitamin C and malic acid. Do not boil — heat releases tannins and makes it bitter. The dried, ground berries (called 'sumac spice') are a staple Middle Eastern seasoning used to add tangy citrus flavor to meats, salads (fattoush), and rice. Birds — including bluebirds, robins, and cedar waxwings — rely heavily on sumac berries as a late-winter emergency food source.

Key Facts

- **ONLY** eat RED-berried sumac — Poison Sumac has WHITE/cream berries and grows in wetlands
- Cold-steep berry clusters for lemonade substitute — never boil (releases bitter tannins)
- Ground dried berries = sumac spice, a prized Middle Eastern seasoning worth \$15+/lb
- Berries persist through winter — critical late-season bird food when little else remains
- Spectacular scarlet fall foliage rivals any ornamental shrub
- Root bark tea used historically as an astringent mouthwash and for fever

Trumpet Creeper



Trumpet Creeper (*Campsis radicans*) is a vigorous native vine that climbs trees, fences, and buildings using aerial rootlets, common throughout Missouri in woodland edges, thickets, and roadsides. The large, pinnately compound leaves (7–11 leaflets) and spectacular clusters of orange-red trumpet-shaped flowers (2–3 inches) bloom July–September. It is one of Missouri's most recognizable summer-blooming vines.

■ Utility & Value

Trumpet Creeper is mildly toxic — contact with the sap and even flowers causes skin irritation and dermatitis in sensitive individuals. The reaction is milder than Poison Ivy but can cause red, itchy rashes, particularly on children. Despite this, it is widely planted as a garden ornamental for its hummingbird attraction — Ruby-throated Hummingbirds are closely associated with this plant and visit the flowers extensively. Its aggressive growth habit requires management in gardens. There is no edible use.

Key Facts

- Causes skin rash in sensitive individuals — handle with care; wear gloves when pruning
- One of the best hummingbird plants in Missouri — Ruby-throated Hummingbirds depend on it
- Aggressive grower — can damage structures and siding with aerial rootlets
- Orange-red trumpet flowers bloom July–September — showy summer color
- Common folk name: 'cow itch vine' — indicating its irritant properties
- Native vine but invasive in garden settings — plant with caution near structures

Water Hemlock



Water Hemlock (*Cicuta maculata*) is widely considered the most violently toxic plant in North America. It grows in wet areas — stream banks, marshes, pond margins, and wet meadows — throughout Missouri. Plants reach 3–6 feet with hollow, chambered stems marked with purple mottling, compound leaves with lance-shaped toothed leaflets, and flat-topped clusters of tiny white flowers (umbels) resembling Queen Anne's Lace. The roots smell faintly of parsnip or carrot and are the most toxic part.

■ Utility & Value

There is absolutely no edible use — Water Hemlock is potentially the most acutely toxic plant in North America. Cicutoxin, concentrated in the roots, causes violent convulsions within 15–60 minutes of ingestion. A piece of root the size of a walnut has killed adult humans. Even touching the sap and then touching the eyes can cause serious harm. It has been mistaken for wild parsnip, water parsley, and other edible umbellifers — this error has killed multiple foragers. When in doubt about any white-flowered member of the carrot family in a wet area, do not eat it. Know this plant's appearance cold.

Key Facts

- MOST TOXIC PLANT IN NORTH AMERICA — a single bite of root can be fatal
- Causes rapid, violent convulsions — death can occur within hours of ingestion
- Hollow, chambered, purple-mottled stem is the key ID feature
- Often mistaken for wild parsnip or water parsley — these errors kill people
- Found in wet areas: stream margins, marshes, wet meadows statewide
- Never forage any white-flowered umbellifers from wet habitats without expert confirmation

Watercress



Watercress (*Nasturtium officinale*) grows in and along cold, clear, spring-fed streams and seeps throughout Missouri, especially in the Ozarks where limestone springs create ideal habitat. The small, round, dark green leaflets float on the water surface on hollow stems. Tiny white flowers appear in spring. Missouri's Ozark region has abundant natural springs, making watercress common in the right habitat. The plant has a peppery, mustard-like bite.

■ Utility & Value

Watercress is one of the most nutrient-dense greens available — the CDC ranks it among the top 5 powerhouse vegetables for nutrient density. High in vitamins K, A, and C, calcium, and glucosinolates (compounds studied for cancer-preventive properties). Eat raw in salads or sandwiches, blend into soup (classic French potage au cresson), or cook briefly. Harvest only from clean, cold, fast-moving spring water — avoid water that may be contaminated by agricultural runoff, as the plant can harbor liver flukes (*Fasciola hepatica*) in water near livestock. Wash thoroughly before eating.

Key Facts

- Only harvest from clear, cold, fast-moving spring water — not slow or agricultural drainage water
- Liver flukes can be present in water near livestock operations — wash and inspect thoroughly
- Ranked among top 5 most nutrient-dense vegetables by the CDC
- Peppery flavor pairs well with eggs, smoked fish, and simple vinaigrettes
- Ozark springs are prime habitat — look along crystal-clear gravel-bottom streams
- Harvest year-round in Missouri's spring-fed streams — watercress is often evergreen

White Snakeroot



White Snakeroot (*Ageratina altissima*) is a native perennial herb, 1–4 feet tall, found in rich, shaded woodlands and forest edges throughout Missouri. The opposite, toothed, heart-shaped leaves and dense flat-topped clusters of tiny white flowers (blooming August–October) can make it superficially attractive. It is the plant responsible for 'milk sickness' — the disease that killed Abraham Lincoln's mother, Nancy Hanks Lincoln, in Indiana in 1818.

■ Utility & Value

White Snakeroot contains tremetol, a complex alcohol that accumulates in the milk and flesh of livestock that eat it. Humans who drink contaminated milk develop 'milk sickness' (trembles) — characterized by weakness, nausea, vomiting, and potentially death. This killed thousands of settlers in the early 1800s before the cause was identified by Dr. Anna Pierce Hobbs Bixby, whose findings were initially dismissed because she was a woman. There is no edible or medicinal use. Keep livestock out of woodland areas where it grows.

Key Facts

- Caused 'milk sickness' epidemic that killed Abraham Lincoln's mother in 1818
- Toxin (tremetol) passes through cow's milk — livestock grazing it contaminate dairy products
- No edible or medicinal use — avoid entirely
- Blooms late August–October when few other woodland plants flower — note and avoid
- Dr. Anna Pierce Hobbs Bixby identified the cause in 1830s but was dismissed by male physicians
- Keep livestock out of shaded woodland areas where this plant grows abundantly

Wild Blackberry & Raspberry



Several *Rubus* species (*R. allegheniensis*, *R. occidentalis*, *R. flagellaris*) colonize Missouri's disturbed areas, old fields, forest edges, roadsides, and clear-cuts statewide. Arching canes bear stout thorns and white 5-petaled flowers in May–June. Blackberries ripen July–August; black raspberries (thimbleberries) ripen slightly earlier in June–July. Black raspberries have a hollow center when picked; blackberries retain a solid white core.

■ Utility & Value

One of Missouri's most abundant wild edibles. Eat fresh, make jam, jelly, cobbler, wine, or vinegar. Berries are high in vitamin C, fiber, and anthocyanins with strong antioxidant and anti-inflammatory properties. Leaves can be dried for tea — historically used for diarrhea and sore throats. Canes provide excellent habitat for songbirds, small mammals, and pollinators. Dense thickets are favored cover for rabbits and white-tailed deer fawns.

Key Facts

- Blackberry has solid white core when picked; raspberry has hollow center
- Peak harvest: black raspberry June–July, blackberry July–August
- First-year canes (primocanes) are vegetative; second-year (floricanes) bear fruit
- Young spring shoots are edible — peel and eat raw or cook like asparagus
- Chiggers are common in blackberry patches — wear long sleeves and tuck pants
- Leaves dried and steeped make an astringent tea for upset stomach and diarrhea

Wild Garlic / Ramps



Wild Garlic and Ramps (*Allium vineale* and *A. tricoccum*) are two distinct but equally prized Missouri edibles. Wild Garlic (field garlic) grows in disturbed ground, lawns, and fields statewide with hollow, tubular leaves and clusters of small bulbs. Ramps (wild leeks) prefer rich, moist, shaded hardwood forests — Ozark hollows and north-facing slopes — producing broad flat leaves in early spring (March–April) that die back before summer. Both emit an unmistakable onion-garlic odor.

■ Utility & Value

Every part of both species is edible — bulbs, leaves, and flowers. Ramps are a springtime delicacy celebrated at festivals across Appalachia and the Ozarks; their flavor is assertive garlic-onion with a hint of leek sweetness. Sauté with butter, scramble with eggs, pickle the bulbs, or use raw in salads and pesto. Wild garlic bulbs can replace cultivated garlic in any recipe. Both are high in vitamin C, vitamin A, and allicin — the antimicrobial compound responsible for garlic's health benefits. Important: always confirm the garlic scent before consuming — toxic Death Camas lacks any onion smell.

Key Facts

- CRITICAL ID: Always crush a leaf and smell for onion/garlic before eating any *Allium* species
- Death Camas resembles ramps but has NO garlic smell — a fatal mistake
- Ramps are a highly seasonal delicacy: harvest window is only 3–4 weeks in early spring
- Sustainable harvest: take no more than 1/3 of a colony; ramps grow very slowly
- Ramp leaves dry and can be used as garlic substitute year-round
- Wild garlic aerial bulbils (top clusters) can be used like minced garlic

Wild Parsnip

Wild Parsnip (*Pastinaca sativa*) is an invasive biennial from Europe, widespread along Missouri roadsides and disturbed areas. Growing 2–5 feet tall, it has flat-topped yellow flower clusters and pinnately compound leaves. The real danger is not ingestion but skin contact: the plant sap contains furocoumarins that cause severe phototoxic burns when skin is exposed to sunlight after contact.

■ Utility & Value

Although the root of cultivated parsnip is edible, wild parsnip contact is extremely hazardous. Furocoumarin in the sap causes phytophotodermatitis — when sap-exposed skin is hit by UV light, severe blistering burns develop within 24–48 hours, leaving dark hyperpigmentation scars that persist for months to years. Never brush against it with bare skin in sunshine. If contact occurs, immediately cover the area from sunlight and wash thoroughly with soap and water. Wear gloves and long sleeves when working near it.

Key Facts

- Sap + sunlight = severe chemical burns — even brief skin contact can be hazardous
- Burns appear 24–48 hours after exposure; dark pigmentation scars persist for months
- Never mow, cut, or clear it with bare skin exposed — use gloves and long sleeves
- Yellow flat-topped flowers distinguish it from white-flowered hemlock relatives
- Common in Missouri roadsides May–July — at peak hazard when blooming
- If contact occurs: cover from sun immediately and wash with soap and water

Wild Strawberry



Wild Strawberry (*Fragaria virginiana*) is a low-growing perennial found in open woods, meadows, roadsides, and forest edges throughout Missouri. The trifoliate leaves have toothed, oval leaflets and prominent veins. White 5-petaled flowers bloom April–May; tiny red berries (1/3–1/2 inch) ripen May–June. While small compared to grocery store berries, the flavor is dramatically more intense — concentrated sweetness far beyond any cultivated variety.

■ Utility & Value

Wild strawberries are among the most flavorful fruits in Missouri — an explosion of sweet-tart flavor in each small berry. All cultivated strawberries are partially descended from this species (crossed with *F. chiloensis* from South America in the 1700s). Eat fresh, make jam, add to salads, or steep in cream for ice cream. Leaves dried make a pleasant, mild tea rich in vitamin C and tannins — used historically for diarrhea and as a mild diuretic. The fruit is high in vitamin C, folate, and anthocyanins. Finding a good patch of wild strawberries is a true foraging prize — their season is brief (2–3 weeks).

Key Facts

- Peak season: mid-May to mid-June — a very short window
- Wild strawberry flavor is 3–5x more intense than cultivated varieties
- All commercial strawberries are part *Fragaria virginiana* in their ancestry
- Leaves dried for tea: mild flavor, high Vitamin C, traditional remedy for diarrhea
- Look in forest edges receiving morning sun and afternoon shade
- Runners spread to form colonies — find one plant, find many nearby



Wood Sorrel (*Oxalis* species — *O. stricta*, *O. violacea*) is a delicate low-growing plant found in woodlands, gardens, lawns, and disturbed areas throughout Missouri. The clover-like leaves are made of three heart-shaped leaflets that fold downward at night or in bright sun. Small yellow (*O. stricta*) or violet (*O. violacea*) 5-petaled flowers appear spring through fall. The plant is easily overlooked but found almost everywhere.

■ Utility & Value

Wood sorrel's bright lemony-sour flavor comes from oxalic acid, making it a pleasant trail snack and salad addition. Leaves, stems, and flowers are all edible raw — their tartness perks up mixed greens and can substitute for lemon in dressings and sauces. The flavor is strongest in spring. Steep in water for a refreshing lemonade-like drink. Like spinach and chard (also high in oxalates), moderate consumption is fine but eating very large quantities is not advised for people prone to kidney stones. Children instinctively eat this plant — it's been called 'sour grass' or 'sourgrass' in folk tradition across Missouri.

Key Facts

- Safe in moderate amounts; high oxalic acid content means large quantities not advised for kidney-stone-prone individuals
- Three heart-shaped leaflets on each stem — similar to clover but taste sour
- Leaves fold shut at night and in heat — a good indicator of evening approaching
- Violet wood sorrel (*O. violacea*) has purple flowers and grows in drier upland areas
- The seed pods explode when ripe, flinging seeds several feet — touch them to see
- Often confused with clover — taste test settles it immediately; clover has no sour taste

Fungi & Mushrooms

13 entries



Golden Chanterelle (*Cantharellus cibarius* and related species) is one of Missouri's most beautiful and flavorful wild mushrooms, fruiting in summer through fall in hardwood forests — particularly under oaks and beech on moist, well-drained slopes. The cap is golden yellow to egg-yolk orange, wavy-edged, and depressed in the center. The undersurface has false gills — forked, blunt-edged ridges that run down the stem — not true knife-sharp gills. The flesh is firm, white, and smells of apricots or fruit.

■ Utility & Value

Chanterelles are among the most commercially valuable and culinarily prized mushrooms in the world, fetching \$15–\$30/pound fresh. The delicate, fruity-peppery-earthly flavor is best highlighted simply: sauté in butter, finish with cream, and serve on toast or pasta. They pair brilliantly with eggs. Dry chanterelles carefully — they don't rehydrate as well as morels, but the dried powder is an incredible seasoning. The Jack-o'-Lantern Mushroom is a dangerous lookalike (true sharp gills, orange color, grows in clusters at wood; Chanterelle has false gills, grows singly from soil).

Key Facts

- False gills (blunt forked ridges) vs. true gills is the key distinction from toxic lookalikes
- Fruit-apricot scent is characteristic — one of the best aroma cues in foraging
- Jack-o'-Lantern lookalike glows faintly blue-green in darkness — chanterelle does not
- Best sautéed in butter — high water content means they steam before browning; use high heat
- Peak season: June–September in Missouri after summer rains
- Fetch \$15–30/lb at markets — premium culinary mushroom worldwide



Chicken of the Woods (*Laetiporus sulphureus* and *L. cincinnatus*) is a spectacular bracket fungus growing in large, overlapping, shelf-like rosettes on dead or dying trees — most commonly oaks, but also cherry, locust, and others — throughout Missouri. The bright orange and yellow colors are unmistakable; no other Missouri mushroom looks like it. Fruiting bodies can reach enormous size — 50+ pounds from a single specimen. *L. cincinnatus* grows from the base/roots and has a white to cream underside; *L. sulphureus* grows higher on the trunk with yellow pores on the underside.

■ Utility & Value

Chicken of the Woods genuinely tastes like chicken — the meaty, firm, fibrous white flesh shreds and absorbs flavors remarkably similarly to chicken breast. Use as a meat substitute in stir-fries, soups, tacos, and pasta dishes; bread and fry for 'chicken' strips. Only harvest young, bright-colored specimens with tender outer edges — old growth becomes tough and dry. Avoid specimens growing on conifers, black locust, or eucalyptus — these can cause reactions. Some people experience mild GI distress on first consumption — start with a small test portion.

Key Facts

- No other Missouri mushroom looks like this — one of the safest wild mushroom IDs
- Only harvest young, tender outer margins — old growth is tough and woody
- Avoid specimens on conifers or black locust — can cause allergic reactions
- Test small amount on first consumption — some individuals experience GI sensitivity
- High in protein, potassium, and antioxidants — a genuinely nutritious meat substitute
- Fruiting season: May–November, with peak in late summer and fall on oaks



The Death Cap (*Amanita phalloides*) is responsible for approximately 90% of mushroom-related deaths worldwide. It was introduced to North America from Europe and has been expanding its range, now documented in Missouri associated with planted oaks and ornamental trees in parks and neighborhoods. The cap is greenish-yellow to olive, the gills are white, the base emerges from a distinctive cup-like volva, and there is a pendant ring (skirt) on the stem. A fully grown Death Cap can kill an adult human with a single cap.

■ Utility & Value

There is no edible use, and no known antidote. Amatoxins (specifically alpha-amanitin) destroy liver and kidney cells. The insidious danger is the symptom timeline: symptoms are delayed 6–24 hours, then apparent recovery occurs for 1–4 days while liver destruction continues silently. By the time jaundice and organ failure appear (day 3–5), massive irreversible damage has occurred. Liver transplant is sometimes the only option. It reportedly tastes pleasant — do not use taste as a safety test. Know the Death Cap by sight. Teach it to your children.

Key Facts

- Responsible for ~90% of mushroom fatalities worldwide
- Symptoms are DELAYED 6–24 hours — by which time severe liver damage has begun
- No antidote exists — treatment is supportive; liver transplant sometimes required
- Volva (cup at base) + ring on stem + white gills + greenish cap = Death Cap
- Reportedly tastes good — taste is NOT a safety test
- Never eat a white-gilled mushroom with a ring and base cup without expert ID



Destroying Angel (*Amanita bisporigera*, *A. virosa*) is a pure white mushroom — cap, gills, stem, ring, and volva all white — found in Missouri's hardwood forests June through October. The pristine white beauty is deeply deceptive. It contains the same lethal amatoxins as the Death Cap and kills with the same delayed mechanism. Button-stage specimens have been mistaken for puffballs (slicing reveals the *Amanita* shape inside), and mature specimens have been confused with button mushrooms.

■ Utility & Value

No edible use. Fatal in the same manner as Death Cap — amatoxin liver and kidney destruction with delayed symptoms and false recovery phase. Young button-stage specimens are occasionally mistaken for Giant Puffballs — this is why slicing puffballs completely in half before eating is non-negotiable. A half-cap of Destroying Angel contains enough toxin to kill an adult. Do not eat any all-white mushroom from the forest floor without expert identification of every feature including the base. The pure white beauty of this mushroom is its danger.

Key Facts

- Half a cap contains enough toxin to kill an adult
- Button stage can be mistaken for puffballs — always slice puffballs to verify pure white interior
- All white: cap, gills, stem, ring, and volva — beauty is its deception
- Same amatoxin and same delayed-symptom mechanism as Death Cap
- Commonly found in Missouri oak and mixed hardwood forests June–October
- No antidote; supportive care only; sometimes requires liver transplant



False Morels (*Gyromitra* species — *G. esculenta*, *G. brunnea*) appear in spring at the same time as true Morels and in similar habitats — making them one of Missouri's most dangerous lookalikes. The cap is irregularly brain-like or saddle-shaped (not a true honeycomb), reddish-brown to dark brown. When cut, the stem is chambered but the cap is not entirely hollow or attached to the stem at its base the way true morels are. *Gyromitra* has a brain-like, wrinkled, lobed cap — *Morchella* has a distinct honeycomb patterned cap.

■ Utility & Value

False Morels contain gyromitrin, which converts to monomethylhydrazine (MMH) in the body — a rocket fuel component and potent toxin causing hemolysis (red blood cell destruction), liver failure, and potentially fatal neurological effects 6–12 hours after consumption. Some people parboil and discard the water to reduce toxin levels — this is NOT reliable and people have died using this method. Cooking volatilizes some toxin but inhalation of cooking vapors is itself dangerous. There is no safe preparation method. Always verify the true honeycomb cap and complete hollow interior of true Morels before eating.

Key Facts

- Spring season overlaps exactly with true Morel — greatest risk of confusion in April–May
- Brain-like wrinkled cap vs. true Morel's honeycomb of pits and ridges
- True Morel is completely hollow when cut; False Morel has chambered, partial interior
- Gyromitrin toxin is related to rocket fuel — not neutralized by parboiling reliably
- Has caused deaths in people who parboiled and ate them believing it was safe
- Even inhaling cooking vapors can cause toxicity — do not cook them indoors



Fly Agaric (*Amanita muscaria*) is the archetypal 'fairy tale' mushroom — bright red to orange cap with white warts (remnants of the universal veil), white gills, white stem, ring, and volva. Found in Missouri in association with pines, spruces, and birches (often ornamental plantings). The white-spotted red cap is one of the most recognized fungi in the world, depicted in fairy tales, video games, and folklore across all cultures for millennia.

■ Utility & Value

Fly Agaric contains ibotenic acid and muscimol — psychoactive compounds (not psilocybin). Effects are unpredictable and include nausea, confusion, delirium, seizures, and coma. It is not reliably psychedelic in any pleasant sense; the experience is typically reported as extremely unpleasant and disorienting. Historical use in Siberian shamanic traditions is documented, but doses are impossible to standardize and toxic reactions are common. The common name comes from European tradition of using it as a fly killer — dissolved in milk, it attracts and kills flies. Do not eat.

Key Facts

- The archetypal red-and-white spotted mushroom of fairy tales and folklore
- Contains ibotenic acid and muscimol — unpredictably toxic, not reliably hallucinogenic
- Named for use as a fly poison — dissolved in milk, the compounds kill insects
- Historical shamanic use documented in Siberia — but dosing is impossible to standardize safely
- White warts on cap are remnants of the egg casing — they can wash off in rain
- In Missouri primarily associated with ornamental pines, spruces, and birch plantings

Galerina marginata



Galerina marginata is a small, inconspicuous brown mushroom growing in clusters on decaying wood — particularly rotting conifer and hardwood logs — throughout Missouri. Cap is brown, bell-shaped to convex, with a ring on the stem. It is easily overlooked as an unimportant 'little brown mushroom' (LBM). This is the danger: it contains the same deadly amatoxins as the Death Cap in comparable concentrations. It has killed foragers who mistook it for edible wood-growing species, particularly Honey Mushrooms or Oyster Mushrooms.

■ Utility & Value

No edible use — fatally toxic. Contains alpha-amanitin at lethal concentrations. It has killed foragers who mistakenly consumed it alongside Honey Mushrooms (*Armillaria* species), which can look similar and grow in the same habitat. The critical rule: do not casually eat any small brown mushroom growing on wood without expert identification. This species represents the risk of foraging by 'shape' rather than detailed feature analysis. Even experienced foragers should be cautious with all wood-growing small brown mushrooms.

Key Facts

- Contains same lethal amatoxins as the Death Cap — in a much smaller, easily overlooked mushroom
- Has killed foragers mixing it with Honey Mushrooms from same log
- Small brown mushrooms (LBMs) should NEVER be eaten without expert identification
- Ring on stem + brown cap + wood-growing + small size = the dangerous profile
- Death timeline same as Death Cap: delayed symptoms, false recovery, organ failure
- Most foraging experts say: if you're not 100% certain of a small brown mushroom, leave it

Giant Puffball



Giant Puffball (*Calvatia gigantea*) is the unmistakable white sphere of Missouri's late summer pastures and woodlands — fruiting bodies range from baseball to basketball to literally beach-ball size (some exceed 50 lbs). Found in meadows, pastures, woodland edges, and lawns after summer rains, particularly August–October. The exterior is white and smooth when young; the interior must be pure white throughout when sliced to be safe.

■ Utility & Value

Giant Puffballs are delicious when the interior is uniformly white — slice like bread, bread and fry, sauté, or grill. They absorb butter and oil beautifully with a mild, earthy flavor. A basketball-sized puffball contains approximately 7 trillion spores and represents more calories than most people could eat in a week. **CRITICAL SAFETY CHECK:** slice in half completely — if white throughout, it's safe. If you see any outline of a developing cap (suggesting an *Amanita* egg inside), discard immediately. Once the interior begins to yellow or turn brown, it has become toxic — only pure white is safe.

Key Facts

- Slice completely in half before cooking — pure white throughout = safe; any color or outline = discard
- Must rule out *Amanita* 'egg' stage inside — slicing reveals it immediately
- Once interior yellows, becomes toxic — only pure white specimens are edible
- A single large specimen can weigh 20+ lbs and feed a family for days
- Dried puffball powder has styptic properties — used to stop minor bleeding historically
- Peak season: August–October, in open grassy areas after substantial rainfall

Hen of the Woods / Maitake



Hen of the Woods (*Grifola frondosa*), called Maitake in Japanese ('dancing mushroom'), is a large polypore mushroom forming enormous rosettes of overlapping gray-brown to tan fronds at the base of oaks — particularly old, large white oaks — in Missouri's forests. A single specimen can weigh 10–100+ pounds. It fruits in September–October, returning to the same tree year after year. The fronds have a wavy, irregular shape and white pores on the underside. It smells pleasantly earthy and mushroomy.

■ Utility & Value

Hen of the Woods is a prized culinary and medicinal mushroom. The flavor is rich, earthy, and deeply savory — excellent roasted, grilled, sautéed, or in soups and ramen. Medicinally, Maitake extracts (particularly beta-glucans and the 'D-Fraction' extract) have been clinically studied for immune system support, blood sugar regulation, and anti-tumor properties. It is sold as an immune supplement for \$30–\$60 per pound dried. Finding one in the wild — especially a large specimen under a prized white oak — is considered one of Missouri's great foraging treasures.

Key Facts

- Returns to the same tree base every fall for decades — mark your spots carefully
- Single specimens can exceed 50 lbs — one find can fill a chest freezer
- Contains beta-glucans shown to enhance immune function and regulate blood sugar
- Sold as 'Maitake extract' supplement for \$30–60/lb — extremely valuable
- Roasting at high heat brings out the best flavor — treat like a meaty vegetable
- Look for large, old white oaks in September–October — most reliable indicator

Jack-o'-Lantern



The Jack-o'-Lantern (*Omphalotus olearius* / *O. illudens*) is a spectacularly bright orange mushroom growing in dense clusters at the base of trees or from buried roots — most commonly oaks — in Missouri. The cap, gills, and stem are all orange. It is famously bioluminescent: the gills glow faint blue-green in complete darkness. It is regularly mistaken for Chanterelles, but the differences are clear on examination.

■ Utility & Value

Jack-o'-Lantern is toxic — causes severe vomiting, diarrhea, and abdominal cramping within 1–3 hours. It is not typically fatal but can cause severe dehydration. The primary confusion is with Chanterelles. Key differences: Jack-o'-Lantern grows in dense clusters from wood/roots (Chanterelle grows singly from soil), has true sharp gills (Chanterelle has false blunt ridges), is uniformly bright orange throughout (Chanterelle has white flesh), and glows in the dark. If you find bright orange mushrooms clustered at a tree base in Missouri, assume Jack-o'-Lantern until proven otherwise.

Key Facts

- Bioluminescent gills glow blue-green in complete darkness — a reliable ID test
- Grows in clusters at wood or buried roots — Chanterelles grow singly from soil
- True sharp gills vs. Chanterelle's false blunt ridges — examine the underside
- Bright orange flesh throughout — Chanterelle has white flesh
- Causes severe vomiting and diarrhea but rarely fatal — very unpleasant experience
- Most commonly found on oaks in Missouri — August through November



Lion's Mane (*Hericium erinaceus*) is one of North America's most remarkable wild mushrooms — a pure white, cascade of soft spines hanging from a central point on dead or dying hardwood trees, primarily oaks and beeches. The entire fungus resembles a white waterfall or a lion's mane, with no cap, gills, or stem — just cascading spine clusters 4–16 inches wide. Found in Missouri's hardwood forests in fall (August–November). No other Missouri mushroom resembles it.

■ Utility & Value

Lion's Mane is prized both as food and medicine. The flavor when sautéed in butter is remarkably similar to crab or lobster — firm, slightly chewy, and seafood-sweet. Medicinally, it is the most well-studied mushroom for neurological health: compounds (hericenones and erinacines) have been shown in clinical trials to stimulate Nerve Growth Factor (NGF) production, potentially supporting memory, focus, and neuroprotection. Studies are exploring its potential in mild cognitive impairment and Alzheimer's research. Sold as a cognitive supplement for \$30–60/lb dried. A dual-use find of genuine medicinal value.

Key Facts

- Stimulates Nerve Growth Factor (NGF) — clinical studies support cognitive benefits
- Flavor resembles crab or lobster when sautéed in butter
- No other Missouri mushroom looks like it — one of the safest IDs in foraging
- Sold as 'Lion's Mane extract' cognitive supplement for \$30–60/lb
- Best harvested young and white — yellowing indicates age and reduced quality
- Found on dying hardwood trees Aug–Nov; favorite hosts are oaks and beeches



True Morels (*Morchella* species — *M. esculenta*, *M. americana*, *M. angusticeps*) are Missouri's most celebrated wild mushroom and the centerpiece of spring foraging culture statewide. The distinctive honeycomb cap — entirely pitted and ridged, hollow throughout the stem, and attached to the stem at its base — makes the Morel the safest large mushroom to identify for beginners. Yellow morels appear first (late March–April), followed by gray and black morels. Peak season is typically when redbuds bloom and soil reaches 50°F.

■ Utility & Value

Morels are among the finest edible fungi in the world — prized by chefs and fetching \$20–\$50+ per pound at farmers markets. The nutty, earthy, deeply umami flavor intensifies dramatically when sautéed in butter. Slice in half, sauté with butter and garlic, deglaze with cream, and serve over pasta or steak. Always cook morels — they can cause GI upset raw. Dry by stringing on thread and hanging — dried morels rehydrate beautifully and retain full flavor. Missouri morel season is a major cultural event — clubs, festivals, and friendly competition define the April landscape. Find them under dying elms, old apple orchards, and south-facing hillsides as the season begins.

Key Facts

- Season: late March–early May — follow redbud bloom and soil temperature as timing guides
- Always hollow when cut lengthwise — solid inside = False Morel (dangerous)
- Found under dying elms, old orchards, cottonwood, and tulip poplar — learn the tree associations
- Cook before eating — raw morels cause GI distress even in the edible species
- Fetch \$20–\$50+/lb — among the most commercially valuable wild mushrooms in Missouri
- Missouri Morel festivals held statewide in April — check local hunting clubs and conservation areas



Oyster Mushrooms (*Pleurotus ostreatus* and *P. pulmonarius*) grow in overlapping clusters on dead and dying deciduous trees — particularly oaks, elms, and beech — throughout Missouri in spring and fall (spring oysters on dying elms are especially productive). The fan-shaped or oyster-shaped caps are white to gray to tan, with white gills running down a short off-center stem. They have a mild, pleasant anise-like odor. One of the most common and easiest wild mushrooms to identify.

■ Utility & Value

Oyster mushrooms are a forager's reliable staple — abundant, easily identified, and excellent eating. Sauté with butter and garlic, add to soups and stir-fries, or roast whole in the oven with olive oil. The texture is silky and tender, flavor mild and subtly briny-umami. High in protein, beta-glucans, and lovastatin (a compound that may reduce cholesterol). Missouri's dying elm trees are the best source — spring emergence after a warm rain following cool nights is the trigger. Store in a paper bag in the refrigerator; avoid plastic which accelerates decay.

Key Facts

- Gills run down the stem (decurrent) — key ID feature
- Anise-like odor when fresh is a helpful confirmation
- One of the best wild mushrooms for beginners — few dangerous lookalikes in Missouri
- Spring fruiting on dying elms is the most reliable Missouri occurrence
- Contains lovastatin — a natural compound that may help reduce cholesterol
- Cultivated commercially; identify wild by same features plus wood-growing habit

Wildlife & Animals

21 entries

Black Widow Spider



The Black Widow (*Latrodectus mactans* — Southern Black Widow) is the second medically significant spider in Missouri, found statewide in undisturbed outdoor areas: woodpiles, outbuildings, under boards, in ground-level debris, and outdoor toilets. The adult female is jet black with a distinctive red hourglass marking on the underside of the spherical abdomen — a pattern unmistakable when visible. Males and juveniles are smaller with different patterns. The web is irregular, strong, and sticky.

■ Utility & Value

Black Widow venom is neurotoxic (alpha-latrotoxin) — it causes latrodectism: severe muscle cramping, rigid abdominal muscles, sweating, and generalized pain. The bite itself may be barely felt initially. Symptoms can be serious — particularly in elderly, children, and immunocompromised individuals — and require medical treatment. Antivenin exists and is effective. Prevention: always wear gloves when working with woodpiles, ground debris, and outbuildings. Check and shake out outdoor furniture, gloves, and equipment that has been left undisturbed. Never reach into dark areas without checking first.

Key Facts

- Red hourglass on underside of black abdomen is definitive ID for adult female
- Neurotoxic venom causes severe muscle cramping and pain — latrodectism
- Bite may barely be felt initially — watch for muscle cramps 30–60 min later
- Found in woodpiles, outbuildings, under boards and debris — always wear gloves
- Antivenin is available and effective — seek medical care for any suspected bite
- Males and immatures are smaller and mottled — less dangerous but also venomous

Black-Legged (Deer) Tick



The Black-legged Tick (*Ixodes scapularis*), commonly called Deer Tick, is the primary vector of Lyme disease in the eastern United States. In Missouri, it is most common in the Ozarks and northern counties with woodland habitat. It is considerably smaller than the Lone Star Tick — nymphs are poppy-seed-sized, adults the size of a sesame seed. The reddish-brown body and dark black legs (especially the scutum area) distinguish it. White-tailed deer and white-footed mice are the primary hosts.

■ Utility & Value

Ixodes scapularis transmits *Borrelia burgdorferi* (Lyme disease), *Anaplasma phagocytophilum* (Anaplasmosis), and *Babesia microti* (Babesiosis). Lyme disease transmission requires 36–48 hours of tick attachment — prompt removal is effective prevention. Early Lyme disease presents with an expanding bull's-eye rash (erythema migrans), fever, and fatigue. Untreated, Lyme can affect joints, heart, and nervous system. Missouri's Lyme disease risk is lower than northeastern states but rising. Check for ticks within 2 hours of leaving tick habitat, paying close attention to scalp, armpits, groin, and behind knees.

Key Facts

- Primary Lyme disease vector — transmission requires 36–48 hours of attachment
- Nymphs are poppy-seed-sized — most common cause of Lyme transmission due to small size
- Bull's-eye rash (erythema migrans) appears 3–30 days after bite — seek medical care
- Use fine-tipped tweezers to remove — grip close to skin and pull steadily upward
- Never twist, crush, or apply heat/petroleum to remove — increases disease transmission risk
- White-footed mouse is the primary disease reservoir — deer are hosts but not transmitters

Brown Recluse Spider



The Brown Recluse (*Loxosceles reclusa*) is one of two medically significant spiders in Missouri and is arguably more common here than anywhere else in the United States. Missouri lies within the heart of its native range. It is 1/4–3/4 inch long, tan to brown, with a distinctive dark brown violin-shaped marking on the cephalothorax (fiddle pointing toward the abdomen). It has 6 eyes in 3 pairs (most spiders have 8). Found in dark, dry, undisturbed spaces: closets, basements, attics, woodpiles, cardboard boxes, and shoes left undisturbed.

■ Utility & Value

Brown Recluse venom is cytotoxic — it destroys cell membranes at the bite site. Most bites cause a bull's-eye lesion that can develop into a necrotic ulcer taking weeks to months to heal. Systemic reactions (hemolysis, kidney failure) are rare but documented.

Prevention: shake out shoes and clothing left in storage, use gloves in woodpiles and basements, keep storage areas tidy. Missouri has the highest density of Brown Recluse populations in the country — understanding them is essential for all Missouri residents. They are reclusive and nonaggressive — most bites occur accidentally when the spider is trapped against skin.

Key Facts

- Missouri has the highest Brown Recluse density in the United States
- Violin marking on back + 6 eyes in 3 pairs = definitive ID (most spiders have 8 eyes)
- Bite can cause a slowly expanding necrotic ulcer taking months to fully heal
- Always shake out shoes, gloves, and clothing stored in basements or attics
- Reclusive — hides in undisturbed dark spaces; bites usually occur when pressed against skin
- Keep sticky traps in basements to monitor populations; reduce clutter to reduce habitat

Bullfrog



The American Bullfrog (*Lithobates catesbeianus*) is North America's largest frog, reaching 3.5–8 inches body length. Found throughout Missouri in ponds, lakes, marshes, and slow river backwaters, it is identified by the enormous tympanum (eardrum circle behind the eye — larger than the eye in males), uniformly green to brownish-green color, and lack of dorsolateral ridges (the raised lines running down the back present in other Missouri frogs). The deep, resonant 'jug-o-rum' call carries far on summer nights.

■ Utility & Value

Bullfrog legs are a prized delicacy — the white, mildly flavored meat is often described as tasting like a cross between chicken and fish. Missouri allows gigging and hand-capture of bullfrogs at night from June 30 through October 31 with a fishing license. Frogging is typically done at night with a bright headlamp — frogs are temporarily blinded by direct light and can be grabbed or gigged. Larger frogs offer more meat; smaller ones are better released. Bullfrog populations can support harvest but are sensitive to water quality — their decline indicates degraded aquatic habitat.

Key Facts

- Missouri frog gigging season: June 30–October 31 (fishing license required)
- Tympanum LARGER than the eye = male; tympanum same size or smaller = female
- Deep 'jug-o-rum' call heard from considerable distances on warm summer nights
- Will eat anything that fits in their mouth: insects, crayfish, mice, small birds, other frogs
- Invasive in western US where introduced — devastating native amphibian populations there
- Hunt by wading shorelines or floating at night with headlamps — eyes reflect bright orange-red

Canada Goose



Canada Goose (*Branta canadensis*) are Missouri's most abundant and recognizable waterfowl — large, black-necked geese with a distinctive white chinstrap patch, brown body, and white undertail. Missouri's resident population (non-migratory 'giant' subspecies) has expanded dramatically since restoration efforts in the mid-20th century and now numbers in the hundreds of thousands statewide. Migratory flocks supplement this in winter. Urban, suburban, and rural populations all thrive.

■ Utility & Value

Canada Goose is excellent table fare when properly prepared — the dark, rich breast meat is best in slow-cooked applications (goose confit, smoked goose, pot pie). Breast meat seared medium-rare is comparable to lean beef. Missouri's goose season is among the most generous in the country due to overabundant resident populations. Special light goose conservation orders run into March. Resident geese can be hunted September 1 onward. Goose fat is an outstanding cooking fat for potatoes, sautéed vegetables, and rendering. Feathers are excellent for down filling.

Key Facts

- Dark breast meat: treat like beef, not like chicken — medium-rare is best
- Goose fat is exceptional for cooking — renders beautifully for potatoes and vegetables
- Resident population hunted September 1 in special early season; migratory season November–January
- Overabundant in urban parks — special urban goose control programs exist in many Missouri cities
- Migrate in distinctive V-formations — lead bird breaks wind for those behind
- Missouri is at the center of the Mississippi Flyway — millions of waterfowl pass through annually

Copperhead



The Copperhead (*Agkistrodon contortrix*) is Missouri's most commonly encountered venomous snake and the cause of the majority of snakebites in the state. Heavily built and 24–40 inches long, it has a distinctive hourglass-banded pattern of chestnut-brown against pale tan — the bands are narrow on the back and wide on the sides, creating the classic pinched-waist hourglass. The copper-colored head is broad and triangular with elliptical pupils and heat-sensing pits between the eye and nostril. Found statewide in rocky, wooded hillsides, brushy areas, old stone walls, and suburban edges.

■ Utility & Value

Copperhead venom is hemotoxic — it destroys tissue and blood cells. Bites are painful and cause significant local swelling, bruising, and necrosis, but are rarely fatal in healthy adults when treated. Most bites occur from accidental stepping on or handling. They are important predators of mice, voles, and frogs. Leave all Copperheads alone — they fill a vital ecological role in rodent control. If bitten: stay calm, immobilize the limb below heart level, remove tight items, call 911. Antivenin (CroFab) is the treatment.

Key Facts

- Most common venomous snakebite in Missouri — responsible for majority of envenomations
- Hourglass bands: narrow on spine, wide on sides — key pattern ID
- Pit organ between eye and nostril detects warm-blooded prey in complete darkness
- Juveniles have bright yellow tail tips used to lure frogs and lizards — 'caudal luring'
- Ambush predator — remains motionless for days waiting for prey to come to it
- If bitten: call 911, keep limb below heart, no tourniquet, no cutting and sucking

Cottonmouth / Water Moccasin



The Cottonmouth (*Agkistrodon piscivorus*) is Missouri's only venomous aquatic snake, restricted to the southern and southeastern portions of the state — primarily the Bootheel lowlands and Ozark border river systems. Heavy-bodied and 24–48 inches, it has a dark olive to black body, often with faint banding visible in younger specimens. The iconic threat display — opening the mouth wide to show the white ('cotton') interior — is unique and unmistakable. The broad, flat-topped head is distinctly triangular and wider than the neck.

■ Utility & Value

Cottonmouth venom is highly hemotoxic — more potent than Copperhead and causing severe tissue destruction. Bites are medical emergencies requiring immediate hospital treatment. Many harmless water snakes (*Nerodia* species) are killed by mistake when mistaken for Cottonmouths — the 'cotton-white mouth display' is the definitive ID. Harmless water snakes open their mouths too but lack the white interior. Cottonmouths control fish, frog, and small mammal populations in wetlands. They are not aggressive toward humans unless cornered or stepped on; back away slowly from any threat-displaying snake.

Key Facts

- White mouth interior (cottonmouth display) is definitive ID — no harmless Missouri snake has this
- Restricted to southern Missouri: Bootheel, Current River, Ozark border river lowlands
- Can strike and envenomate underwater — unique among Missouri venomous snakes
- Dark body, flat-topped wide triangular head wider than neck
- Many harmless *Nerodia* water snakes are killed by misidentification — learn both well
- Semiaquatic — found basking on logs over water, swimming in swamps and river margins

Crayfish



Missouri hosts over 35 native crayfish species — more than almost any other state — making it a global hotspot for crayfish diversity. The most common and familiar is the Virile Crayfish (*Faxonius virilis*) and related Ozark endemics found in clear streams. Crayfish are 1.5–6 inches, lobster-shaped with prominent claws, hard exoskeleton, and 10 legs. They are found under rocks in virtually every Missouri stream, spring, and pond, and are a critically important prey species for bass, catfish, herons, raccoons, and otters.

■ Utility & Value

Crayfish are edible and delicious — the tail meat is sweet and shrimp-like; the claws of large individuals are also meatful. Boil in seasoned water (Cajun spice, salt, garlic) for 5–7 minutes and eat the tail. Missouri allows crayfish to be taken by hand or with traps for personal use (check current MDC regulations). They are also the single best live bait for Ozark Smallmouth Bass, Largemouth Bass, and Channel Catfish. The Ozark Hellgrammite and crayfish used as live bait produce consistently larger catches than artificial lures in Missouri's clearest streams.

Key Facts

- Missouri has 35+ native crayfish species — a global biodiversity hotspot
- Best live bait for smallmouth bass, catfish, and large bluegill in Missouri streams
- Boil with Cajun seasoning for 5–7 minutes — tail meat is sweet and shrimp-flavored
- Shed (molt) their exoskeleton to grow — soft-shell crayfish are irresistible bass bait
- Several Missouri endemic species are federally threatened — do not collect rare species
- Find them by lifting flat rocks in streams — they dart backward when disturbed

Eastern Cottontail



The Eastern Cottontail (*Sylvilagus floridanus*) is Missouri's most abundant small game animal and the most hunted rabbit in North America. Found statewide in brushy thickets, woodland edges, old fields, and suburban areas, it is 14–17 inches long with the distinctive white powder-puff tail that gave it its name. Cottontails are crepuscular — most active at dawn and dusk — and spend daylight hours in 'forms' (shallow depressions in dense cover).

■ Utility & Value

Cottontail is excellent table fare — tender, mild white meat that takes seasoning well. Fry, stew, or braise. Missouri's rabbit season runs October–February. Beware of tularemia (rabbit fever, *Francisella tularensis*) — wear rubber gloves when field dressing; cook thoroughly to 165°F. Tularemia risk is highest in late summer; fall/winter rabbits after the first hard freeze are lower risk. Cottontails also support a rich predator community: great horned owls, red-tailed hawks, foxes, coyotes, mink, and bobcats all depend on them as a primary prey species.

Key Facts

- Missouri's most hunted small game animal — over 1 million harvested in good years
- Wear gloves when field dressing — tularemia (rabbit fever) is a real risk, cook to 165°F
- Tularemia risk lowest after first hard freeze in November
- Females (does) can produce 3–5 litters per year of 4–6 young — explosive population potential
- Snowshoe Hare (*S. americanus*) is a separate rarer Missouri species in boreal forest edges
- Best habitat: brushy thickets with escape cover within 30 feet of feeding areas

Feral Hog



Feral Hogs (*Sus scrofa*) are an invasive non-native species causing significant agricultural and ecological damage across Missouri. They range 100–400+ pounds with coarse dark hair, a straight tail (unlike domestic pigs' curly tail), a long mobile snout, and — on boars — large curved tusks (upper and lower canines that continuously sharpen against each other). Missouri populations are concentrated in the Ozarks and southwestern counties but are expanding rapidly. They travel in family groups (sounders) of 6–30 animals.

■ Utility & Value

Feral hog meat is edible and can be excellent — leaner and more flavorful than domestic pork when the animal is young. Younger hogs under 100 lbs have the best meat quality. Older boars can have strong odor and require long, moist cooking. Always test for *Trichinella* and cook to 160°F internal temperature. Missouri has no closed season or bag limit for feral hogs — landowners and hunters are encouraged to eliminate them on sight. They root up native plant communities, destroy turkey and deer nesting sites, contaminate water sources, and cause millions in agricultural crop losses annually in Missouri.

Key Facts

- No closed season or bag limit in Missouri — actively encouraged to hunt and remove
- Cause estimated \$1.5 billion in agricultural damage annually in the US
- Straight tail (not curled) and large tusks on boars are key distinguishing features
- Rooting destroys native plant communities and creates erosion — invasive ecological disaster
- Always cook to 160°F internal temp — carry *Trichinella* and *Brucella*
- Sows can reproduce at 6 months and produce 2 litters/year of 4–8 piglets — eliminate quickly

Gray & Fox Squirrel



Missouri hosts two native tree squirrels: the Gray Squirrel (*Sciurus carolinensis*) and the Fox Squirrel (*Sciurus niger*). Gray Squirrels are smaller (12–20 inches), salt-and-pepper gray, and prefer dense mature forest interior — Ozark hardwoods and river bottoms. Fox Squirrels are larger (18–26 inches), rusty-orange below, brownish above, and prefer more open, edge habitats — rural farmland with woodlots, open oak savannas. Both species cache mast (nuts and acorns) in fall and rely on spatial memory to relocate caches.

■ Utility & Value

Squirrel is Missouri's most overlooked but most sustainable wild game protein — populations support heavy harvest with no decline. The meat is rich, sweet, and nutty-flavored (from their mast diet), best suited to slow-cooking in stew, Brunswick stew, or braised in gravy. Missouri's squirrel season opens in late May and runs through January. Young squirrels (June–August juveniles) can be fried like chicken; older animals need braising. Squirrels are critically important seed dispersers — their imperfect nut-caching habits are responsible for planting millions of oaks and hickories annually.

Key Facts

- Missouri's most sustainable small game — populations withstand heavy harvest
- Rich, flavorful meat best in Brunswick stew or braised dishes
- Season opens late May — one of Missouri's earliest small game seasons
- Fox Squirrel prefers open woodlots; Gray Squirrel prefers dense mature forest interior
- A single squirrel may cache 10,000 nuts per fall — critical forest regeneration role
- Listen for 'rain of acorns' from above — feeding squirrel sound on a calm day

Lone Star Tick



The Lone Star Tick (*Amblyomma americanum*) is Missouri's most abundant and aggressive tick species, found statewide in woodlands, brushy areas, and grassy edges. The female is easily identified by a single white dot (the 'lone star') in the center of the brown scutum. All three life stages (larvae, nymphs, adults) actively seek hosts by questing — climbing vegetation and reaching with their front legs. Larvae ('seed ticks') can aggregate in the hundreds on a single host.

■ Utility & Value

The Lone Star Tick transmits several pathogens: *Ehrlichia chaffeensis* (Human Monocytic Ehrlichiosis), *Ehrlichia ewingii*, Heartland virus, Bourbon virus, and STARI (Southern Tick-Associated Rash Illness). Additionally, its bite can induce alpha-gal syndrome — a potentially lifelong allergy to red meat (mammalian meat) caused by the tick's alpha-gal molecule triggering antibody production. Alpha-gal syndrome has been increasingly diagnosed across Missouri and can cause severe anaphylaxis to beef, pork, and venison. Use DEET or permethrin, check thoroughly after outdoor activities.

Key Facts

- Can cause alpha-gal syndrome — a potentially permanent allergy to ALL red meat
- Single white dot on female's back — most distinctive Missouri tick
- Transmits *Ehrlichia* and Heartland virus — both serious bacterial/viral illnesses
- All three life stages (larva, nymph, adult) aggressively seek hosts
- Most active April–August; larvae (seed tick masses) peak in midsummer
- Use permethrin on clothing (not skin) for multi-day prevention; DEET on skin

Mallard Duck



The Mallard (*Anas platyrhynchos*) is North America's most abundant and recognizable duck and the most important waterfowl species in Missouri. The male (drake) has an iridescent green head, white collar, chestnut breast, and yellow bill. The female (hen) is mottled brown with an orange-sided bill. Both sexes show a blue-purple speculum (wing patch) bordered by white. Found on virtually every body of water in Missouri — from the smallest farm pond to major river impoundments.

■ Utility & Value

Mallard is the duck of choice for waterfowl hunters and the basis for commercial duck production (all domestic duck breeds derive from Mallard). The dark, rich breast meat is superb — best seared medium-rare, slow roasted, or as duck confit. Missouri's duck season runs approximately November–January; the Bootheel's rice fields and Mississippi River corridor are among the most productive duck hunting areas in North America. Mallard feathers (particularly drake plumage) are prized for fly tying. Rendered duck fat is an outstanding cooking fat.

Key Facts

- All domestic duck breeds (Pekin, Rouen, etc.) are descendants of wild Mallards
- Missouri's Bootheel and Mississippi River corridor are world-class waterfowl areas
- Breast meat best medium-rare — overcooking makes it tough and livery
- Dabbling duck: tips up to feed; doesn't dive like Canvasback or Ring-neck
- Peak migration: November through early December on the Mississippi Flyway
- Mallard leg bands contribute critical data to waterfowl management — report banded birds to USFWS

Raccoon



The Raccoon (*Procyon lotor*) is one of Missouri's most adaptable and intelligent wild mammals, thriving in every habitat from dense Ozark forest to downtown Kansas City. The distinctive black facial mask and banded tail are unmistakable. Raccoons are omnivorous opportunists — crayfish, corn, eggs, fruit, garbage, pet food, and nearly anything else is on the menu. They use their dexterous front paws to manipulate food and open containers with remarkable precision. Nocturnal but commonly seen at dusk.

■ Utility & Value

Raccoon is a traditional Missouri wild game meat — particularly in the rural Ozark tradition. The dark, rich meat is best parboiled to tenderize, then roasted or braised with sweet potatoes and onions. Raccoon pelts were historically Missouri's most valuable fur and remain commercially traded. Raccoon hunting with hounds is a deep Missouri cultural tradition — particularly nighttime treeing with coonhounds. Raccoons can carry rabies and roundworm (*Baylisascaris procyonis*) in feces — never handle dead raccoons without gloves; their feces are a genuine health hazard particularly for children.

Key Facts

- Raccoon roundworm (*Baylisascaris*) in feces can cause serious neurological disease — never let children play where raccoons defecate
- Coonhound hunting (treeing) is a deeply rooted Missouri outdoor tradition
- Raccoon fat was traditionally rendered for cooking and waterproofing leather
- Dexterous hands can open door latches, cooler clasps, and childproof containers
- Missouri's raccoon pelt is commercially harvested — check current MDC trapping regulations
- Carry rabies, distemper, and parvovirus — do not approach disoriented daytime raccoons

Snapping Turtle



The Common Snapping Turtle (*Chelydra serpentina*) is Missouri's largest aquatic turtle, reaching 8–18 inches shell length and weighing up to 35+ pounds. Found in virtually every permanent body of water in Missouri — ponds, lakes, rivers, swamps, and drainage ditches — it is an apex predator in aquatic ecosystems. The rough, keeled carapace is often algae-covered; the tail is thick and dinosaur-like with saw-tooth dorsal scales. The neck is extremely long and can reach nearly any point on the shell.

■ Utility & Value

Snapping Turtle is a prized wild protein source in Missouri's rural tradition. The meat is white, mild, and sweet — often described as having 'seven flavors' (different muscle groups taste distinctly different). Traditional preparation: turtle stew, turtle soup, or fried snapping turtle. Missouri allows harvest with a fishing license during prescribed seasons. Snapping turtles regulate aquatic prey populations including frogs, fish, waterfowl, and carrion. Never pick up by the tail — it can injure the spine; use a stick to guide into a bucket or grasp the shell at the rear quarters.

Key Facts

- Missouri's largest aquatic turtle — can exceed 35 lbs in old specimens
- Seven distinct muscle groups each taste different — prized by wild game cooks
- Harvest with Missouri fishing license; check current season regulations at mdc.mo.gov
- Neck extends nearly the full length of the shell — bite range is much longer than expected
- Never pick up by tail — causes spinal injury to the turtle; grasp rear of shell
- Can survive months without food and breathe underwater via cloacal bursae in winter

Snapping Turtle (Table)



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Timber Rattlesnake



The Timber Rattlesnake (*Crotalus horridus*) is Missouri's largest venomous snake, reaching 36–60+ inches. It occurs in rocky, forested Ozark hillsides, cedar glades, and bluff bases primarily in southern Missouri. Two color phases exist: a yellow phase with dark chevron-banded pattern on yellow-gray background, and a dark (melanistic) phase that appears nearly black. The rattle is unmistakable when audible. A thick-bodied, heavily built snake with a broad triangular head and heat-sensing pits.

■ Utility & Value

Timber Rattlesnake venom is a complex hemotoxic-neurotoxic mixture — bites are serious medical emergencies with potentially fatal outcomes without antivenin treatment. Despite their size and potency, Timber Rattlesnakes are generally docile unless provoked and rely on their cryptic camouflage as primary defense. They are a Species of Conservation Concern in Missouri due to habitat loss and collection. They are critically important predators of small mammals — a single rattlesnake may consume dozens of rodents annually, providing significant agricultural and ecological services.

Key Facts

- Missouri's largest venomous snake — thick-bodied and powerfully built
- Species of Conservation Concern — do not collect or harm
- Rattle adds a segment each time snake sheds, not one per year
- Rocky, south-facing Ozark hillsides and cedar glades are prime habitat
- Ambush predator: waits motionless at rodent runways for days
- Can sense body heat via pit organs — can strike accurately in complete darkness

Virginia Opossum



The Virginia Opossum (*Didelphis virginiana*) is North America's only native marsupial and one of Missouri's most misunderstood mammals. Found statewide in forests, farmland, and suburban areas, it is the size of a large house cat with a pointed snout, naked prehensile tail, opposable rear thumbs, and 50 teeth — more than any other North American land mammal. Females carry young in a fur-lined pouch (marsupium) after an 13-day gestation — the shortest of any North American mammal.

■ Utility & Value

Opossum is edible and historically eaten throughout the rural South and Ozarks — 'possum and sweet potatoes' was a classic Depression-era Missouri dish. The meat is fatty and mild. They are extraordinary tick predators — a single opossum can kill and eat 5,000+ ticks per season through grooming, providing massive ecological disease control services. They are also immune to most snake venoms including Copperhead and Rattlesnake — peptides derived from opossum blood are being studied as potential antivenin treatments. 'Playing possum' (tonic immobility) is an involuntary catatonic state, not a choice.

Key Facts

- Consumes 5,000+ ticks per season through grooming — one of the best natural tick controllers
- Immune to most pit viper venoms — being studied for antivenin research
- Playing dead ('playing possum') is involuntary tonic immobility — they cannot control it
- North America's only marsupial — young born the size of a honeybee and crawl to the pouch
- 50 teeth — more than any other North American land mammal
- Rarely carry rabies — low body temperature makes survival of the virus difficult

Western Pygmy Rattlesnake



The Western Pygmy Rattlesnake (*Sistrurus miliarius streckeri*) is Missouri's smallest venomous snake, typically 15–22 inches, found primarily in the Ozark Highlands of southern Missouri in rocky glades, open woodlands, and hillside habitats. Its small size and gray-patterned body with dark blotches make it easy to overlook. The rattle is so small it sounds more like a buzzing insect than a classic rattlesnake rattle — an important safety note for outdoor workers.

■ Utility & Value

Despite its small size, the Pygmy Rattlesnake's venom is medically significant — bites cause painful local tissue damage, swelling, and necrosis. Fatalities are rare but documented. The small rattle sound means many people don't recognize the warning. It feeds on mice, lizards, and frogs, filling an important Ozark ecosystem role. A Species of Conservation Concern in Missouri — their populations are locally declining from habitat loss. Never attempt to handle or kill — observe and move away.

Key Facts

- Missouri's smallest venomous snake — easy to overlook due to size
- Rattle sounds like a buzzing insect, not a classic rattle — often unrecognized
- Gray body with dark dorsal blotches and rusty-red stripe along spine
- Primarily in Ozark Highlands — rocky glades and open cedar woodlands
- Species of Conservation Concern — declining in Missouri
- Despite small size, bites cause significant local tissue damage

White-Tailed Deer



White-tailed Deer (*Odocoileus virginianus*) are Missouri's most important big game species and among the most abundant large mammals in North America. Missouri hosts one of the densest and trophy-quality whitetail populations in the country. Bucks grow antlers annually — velvet covers the antlers April–August as they grow, then is shed in September when bucks harden antlers for the rut. The rut peaks in mid-November in Missouri, when bucks abandon caution and move extensively during daylight hours.

■ Utility & Value

Whitetail venison is among the leanest, most nutritious red meats available — high protein, low fat, rich in iron and B vitamins. Missouri's archery season opens October 1, firearms season in November, and antlerless seasons extend into January. The Missouri record typical antler score exceeds 200 inches. Deer also provide hide (leather, buckskin), bone (tools), sinew (cordage), and fat (tallow for cooking and candles). Annual vehicle collisions with deer cost Missouri hundreds of millions annually — awareness peaks in October–November rut.

Key Facts

- Missouri's #1 big game animal — over 300,000 harvested annually
- Rut peaks mid-November — bucks abandon caution and move all day
- Velvet shed: August–September; rut sign (rubs, scrapes) begins October
- Whitetail venison is leaner than chicken breast and richer in iron than beef
- Fawns born May–June; spotted coat provides camouflage in dappled forest light
- Home range: 1–3 square miles for does; bucks may roam 5+ miles during rut



Wild Turkey (*Meleagris gallopavo*) are a Missouri conservation success story — once nearly eliminated statewide by market hunting and habitat loss, they now number over 400,000 birds after Missouri's landmark restoration program beginning in the 1950s. The largest native North American game bird, adult toms reach 20–25 lbs with beard (modified chest feathers), sharp leg spurs, and the full fan-tail display used in spring courtship. The distinctive 'gobble' of spring toms carries over a mile in still air.

■ Utility & Value

Wild turkey is some of the finest wild game table fare available in Missouri — the dark, rich breast meat is far superior in flavor to commercial turkey. Missouri's spring gobbler season runs April–May (the peak of the rut when toms are vocal and actively responding to calls). Fall seasons for both sexes also exist. Wild turkey feathers were used by Native peoples for fletching arrows and ceremonial regalia. The drumstick meat can be slow-cooked for hours to tenderize. Young birds (jakes and jennies) have the most tender meat; old toms are best braised.

Key Facts

- Missouri turkey population: 400,000+ birds after landmark MDC restoration program
- Spring gobbler season: April–May — peak gobbling activity is early morning
- Tom turkey identifies by: beard, leg spurs (length indicates age), bright head colors in spring
- Wild turkey can fly 55 mph in short bursts and run 25 mph
- Outstanding eyesight: 270-degree field of view; can see color; nearly impossible to stalk
- Mast (acorn, hickory) is the primary fall/winter food — find mast, find turkeys

Fish

4 entries

Channel Catfish



The Channel Catfish (*Ictalurus punctatus*) is Missouri's most popular sport fish and the state fish of Missouri. Found in virtually every river, stream, reservoir, and lake in the state, it is identified by its deeply forked tail, spotted sides (on young fish), and 8 whisker-like barbels around the mouth. It reaches 10–30+ pounds in Missouri's major rivers. The Missouri and Mississippi Rivers produce trophy-class fish.

■ Utility & Value

Channel Catfish are superb table fare — firm, mild, white-fleshed and adaptable to frying, baking, grilling, or smoking. Missouri's catfish fry culture is a deeply rooted tradition. They are scavengers and predators, attracted to cut bait, chicken liver, stink bait, and prepared dough baits. The most productive Missouri catfishing is during the spawn (May–June when males guard nest sites), nighttime in summer, and deep holes in winter. The spines on the dorsal and pectoral fins can puncture skin deeply — handle with care; 'stinging' is common for beginners.

Key Facts

- Missouri's official State Fish
- Fin spines can puncture deeply and become infected — handle catfish from behind the pectoral fins
- Spawn May–June — males guard nests aggressively; don't disturb nesting areas
- Best fishing: night, warm water, near structure and holes in rivers
- Stink bait and chicken liver are traditional Missouri channel cat baits
- Blue Catfish (*I. furcatus*) grows larger — Missouri record exceeds 100 lbs in the Missouri River

Crappie



Black Crappie (*Pomoxis nigromaculatus*) and White Crappie (*P. annularis*) are Missouri's premier panfish and among the finest-tasting freshwater fish in North America. Both species are 6–14 inches, with the distinctive compressed, deep body and irregular dark speckled pattern (Black Crappie has randomly scattered spots; White Crappie has more vertical barring). Both inhabit clear lakes, reservoirs, and slow rivers, congregating near structure — brush piles, dock pilings, fallen trees, and rock piles.

■ Utility & Value

Crappie are widely considered the best-tasting freshwater fish in Missouri — the white, sweet, flaky fillets are superb for frying, and a productive crappie hole can produce dozens of fish in a single outing. Spring spawning (April–May) is the prime fishing season when fish move into shallows around brush and dock structures. White and ultralight jigs in 1/32 to 1/8 oz sizes are the classic Missouri approach. 'Spider rigging' — using multiple long rods off the bow of a slow-moving boat — is a Missouri tradition for covering water and finding schools.

Key Facts

- Often rated the best-tasting freshwater fish in Missouri among experienced anglers
- School fish — if you catch one, slow down and work the area thoroughly
- Spawn April–May in 55–65°F water near shallow brush and woody structure
- Black Crappie: random spots, prefers clearer water; White Crappie: vertical bars, tolerates turbidity
- Nocturnal feeding in summer — nighttime dock fishing under lights is extremely productive
- Missouri has no minimum length limit on crappie in most waters — check MDC regulations

Wildlife & Animals **Largemouth Bass**



Largemouth Bass (*Micropterus salmoides*) is the most popular sport fish in the United States and a fixture of Missouri's reservoirs, lakes, ponds, and slow-moving rivers. It is identified by a dark lateral stripe along the side, a jaw that extends past the rear of the eye (the defining feature vs. Smallmouth Bass), and a nearly divided dorsal fin. Missouri's major reservoirs — Lake of the Ozarks, Truman, Table Rock, Stockton, and Bull Shoals — support world-class largemouth populations.

■ Utility & Value

Largemouth Bass are catch-and-release in tournament culture, but they are also excellent table fare when small to medium (1–3 lbs) — the white, flaky meat is mild and sweet. Large fish (4+ lbs) are better released for the fishery. They are ambush predators striking from cover — target weed edges, dock shadows, laydowns, and points. Missouri's bass fishing peaks in spring (March–May pre-spawn and spawn period) and fall (September–October as fish feed aggressively before winter). Table Rock Lake consistently ranks among the top largemouth fisheries in the country.

Key Facts

- Jaw extends PAST the rear of the eye — distinguishes it from Smallmouth and Spotted Bass
- Spawn in April–June when water reaches 60–65°F — male guards nest; don't target nesters
- World record: 22 lbs 4 oz (1932, Georgia) — Missouri state record over 13 lbs
- Most responsive to lures in water 60–75°F — spring and fall peak seasons
- Table Rock Lake and Lake of the Ozarks are nationally recognized largemouth destinations
- Best eating size: 12–15 inches; large fish best released to maintain quality fishery

Wildlife & Animals **Ozark Smallmouth Bass**



Smallmouth Bass (*Micropterus dolomieu*) reach their finest expression in Missouri's Ozark clear-water streams — the Current, Jacks Fork, Meramec, Gasconade, and Niangua Rivers. The Ozark Smallmouth is widely regarded as the most acrobatic, pound-for-pound hardest-fighting freshwater fish in Missouri. Bronze to greenish-brown with vertical dark bars and a jaw that does NOT extend past the rear of the eye (vs. Largemouth). Red eyes are a useful ID feature.

■ Utility & Value

Ozark smallmouth float trips are a Missouri tradition rivaling deer hunting and turkey hunting as a cultural institution. The Current River, Jacks Fork, and Meramec offer world-class float fishing for smallmouth from late May through September. Crayfish, hellgrammites, and topwater lures fished near gravel bars and bluff bases are classic approaches. Smallmouth are excellent table fare but the catch-and-release ethic is strongly encouraged on Ozark streams to preserve quality. The Ozark National Scenic Riverways protects much of the premier smallmouth habitat.

Key Facts

- Jaw does NOT extend past rear of eye — key difference from Largemouth Bass
- Ozark float trip tradition: canoe/kayak with light tackle, mid-May through September
- Current River and Jacks Fork (both in Ozark NSR) are national smallmouth destinations
- Water must be clear and cool — smallmouth decline in turbid, warm water
- Crayfish is the #1 smallmouth food item in Ozark streams; match with brown soft plastics
- Catch-and-release ethic strongly encouraged to maintain quality wild populations

Animal Tracking

14 entries

Animal Tracking **American Beaver**



American Beaver (*Castor canadensis*) tracks are large, 5-toed, with the distinctive webbing between hind toes creating a broad, flat hind track 4–6 inches long that resembles a small human hand with webbing. The flat, paddle-shaped tail frequently drags in mud or snow, obliterating the track pattern between sets. The most reliable beaver sign is not the track itself but the chewing evidence — stumps with characteristic pointed, pencil-sharpened cuts from gnawing, mud-plastered dams, lodges, and food caches (sticks anchored in stream bottom near the lodge).

■ Utility & Value

Beaver are Missouri's largest native rodent and among the most ecologically important mammals in the state. Their dam-building creates ponds that support Wood Ducks, Herons, otters, mink, and amphibians. Missouri allows beaver trapping for commercial fur (prime December–February). Beaver castor (the musky gland secretion) is a highly effective trapping lure for multiple fur bearer species. The meat is edible — rich, dark red, and similar to venison; the tail is traditionally slow-cooked as a rendered fat. Some Missouri counties pay bounty for beaver control on agricultural land.

Key Facts

- Hind track: 4–6 inch, 5-toed with webbing; tail drag often obliterates trail
- Chewing sign: pointed, pencil-sharpened tree stumps — most reliable beaver indicator
- Fresh gnawing (pale wood, no weathering) indicates active beaver within days
- Mud dam construction creates ponds supporting dozens of other wildlife species
- Castor (gland secretion): one of the most effective predator trapping lures in Missouri
- Missouri's largest native rodent; adults reach 40–60 lbs in some older individuals

Animal Tracking **American Black Bear**



American Black Bear (*Ursus americanus*) has returned to Missouri's Ozarks through natural recolonization from Arkansas, with sightings confirmed in Missouri's southern counties. The tracks are large (front 3–4 inches wide, hind 6–7 inches long) with 5 toes and prominent claw marks 1–1.5 inches in front of each toe. The hind track resembles a large, wide, flat human foot. Claw marks on smooth-barked trees (beech, aspen) are a classic sign; bear scat (large, tubular, full of berry seeds and acorn fragments) is found along travel routes.

■ Utility & Value

Missouri currently has no bear hunting season — Black Bear are protected as a recovering species. Report any bear sign, tracks, or sightings to MDC immediately — it contributes to population monitoring of this significant wildlife recovery. Black Bears are not typically dangerous in Missouri's current context (low population density, abundant food) but proper food storage practices should be employed in Ozark camping areas. Their ecological role as seed dispersers (eating and depositing fruit seeds) and soil turners (digging for insects and roots) makes them a keystone species.

Key Facts

- Missouri's recovering population: recolonizing from Arkansas; now confirmed in southern counties
- No hunting season — fully protected in Missouri currently; report sightings to MDC
- Front track: 3–4 inch wide; hind track: 6–7 inch, human-foot-shaped
- Claw marks on smooth trees (tooth-marks climbing) — look for claw-scored tree bark at 6+ feet
- Scat: large tubular, berry seeds and acorn fragments when in season; 1.5–2.5 inch diameter
- Not dangerous but practice proper camp food storage in southern Missouri Ozarks

Animal Tracking **Bobcat**



Bobcat tracks are round, 1.5–2.5 inches, with 4 toes and NO claw marks — the retractile claws are held sheathed while walking, a diagnostic difference from all canid tracks. The heel pad has a distinctive 3-lobed leading edge (2 lobes on the trailing edge). The stride is typically 10–16 inches. The Bobcat's stalk-and-ambush hunting style creates irregular, slow, cautious track patterns near thick brush, fallen logs, and rock piles — very different from the purposeful direct-register trot of foxes and coyotes.

■ Utility & Value

Bobcat is Missouri's only wild cat and a Species of Conservation Concern in some regions. Missouri has a regulated bobcat trapping season with limited permit numbers. Prime bobcat habitat — rocky Ozark terrain with dense cover — is identifiable by their tracks, scrapes, and scat deposited on prominent boulders and logs. Finding and reading bobcat sign is one of Missouri's more challenging and rewarding tracking experiences. They prey primarily on rabbits and squirrels but are capable of taking wild turkey and deer fawns.

Key Facts

- Track: round, 4 toes, NO claw marks (retractile claws) — the defining feline track feature
- Heel pad: 3-lobed leading edge — key to confirming bobcat vs. large domestic cat
- Domestic cat tracks are smaller (1–1.5 inch) and rarely found far from human habitation
- Scrapes of leaves with scat deposited on top at trail junctions = bobcat scent post
- Irregular slow-stalk trail pattern near cover vs. coyote's purposeful straight-line trot
- Missouri's only native wild cat — Ozark rocky terrain with thick cover is prime habitat

Animal Tracking **Coyote**



Coyote tracks are oval, 2–3 inches long, with 4 toes showing claws (claws always register in coyote tracks — a key distinction from bobcat). The negative space between the pads forms a distinctive X pattern. The track is similar to a medium dog but narrower and more oval; the coyote's efficient direct register trot (hind foot lands in front foot print) creates a clean, nearly single-file trail unlike the sloppy, overlapping dog trail. Scent posts (urine on prominent rocks, grass clumps, logs) mark territory.

■ Utility & Value

Coyotes are Missouri's most important and adaptable predator — controlling rabbits, rodents, and fawn populations. They are also a popular fur bearer and calling target. Electronic predator calls simulating rabbit distress or fawn bleats are highly effective January–March. Coyote fur has moderate market value. Their tracks along stream banks and field edges indicate travel routes used for food and territory marking. Understanding coyote behavior helps hunters and livestock producers — coyotes rarely attack healthy adult livestock but can take lambs, goats, and poultry.

Key Facts

- Track: 4 toes with claws always showing; oval, 2–3 inches; X pattern in negative space
- Direct register trot: hind foot lands precisely in front foot track — nearly single-file trail
- Claws always register in tracks — key distinction from Bobcat (which retracts claws)
- Scent posts at prominent objects along regular routes; urine has strong musky odor
- Scat often twisted with hair and bone fragments; deposited at trail intersections
- Peak calling success: January–March when pups not yet present and food is scarcer

Animal Tracking **Eastern Cottontail**



The Eastern Cottontail (*Sylvilagus floridanus*) is Missouri's most abundant small game animal and the most hunted rabbit in North America. Found statewide in brushy thickets, woodland edges, old fields, and suburban areas, it is 14–17 inches long with the distinctive white powder-puff tail that gave it its name. Cottontails are crepuscular — most active at dawn and dusk — and spend daylight hours in 'forms' (shallow depressions in dense cover).

■ Utility & Value

Cottontail is excellent table fare — tender, mild white meat that takes seasoning well. Fry, stew, or braise. Missouri's rabbit season runs October–February. Beware of tularemia (rabbit fever, *Francisella tularensis*) — wear rubber gloves when field dressing; cook thoroughly to 165°F. Tularemia risk is highest in late summer; fall/winter rabbits after the first hard freeze are lower risk. Cottontails also support a rich predator community: great horned owls, red-tailed hawks, foxes, coyotes, mink, and bobcats all depend on them as a primary prey species.

Key Facts

- Missouri's most hunted small game animal — over 1 million harvested in good years
- Wear gloves when field dressing — tularemia (rabbit fever) is a real risk, cook to 165°F
- Tularemia risk lowest after first hard freeze in November
- Females (does) can produce 3–5 litters per year of 4–6 young — explosive population potential
- Snowshoe Hare (*S. americanus*) is a separate rarer Missouri species in boreal forest edges
- Best habitat: brushy thickets with escape cover within 30 feet of feeding areas

Animal Tracking **Feral Hog**



Feral Hogs (*Sus scrofa*) are an invasive non-native species causing significant agricultural and ecological damage across Missouri. They range 100–400+ pounds with coarse dark hair, a straight tail (unlike domestic pigs' curly tail), a long mobile snout, and — on boars — large curved tusks (upper and lower canines that continuously sharpen against each other). Missouri populations are concentrated in the Ozarks and southwestern counties but are expanding rapidly. They travel in family groups (sounders) of 6–30 animals.

■ Utility & Value

Feral hog meat is edible and can be excellent — leaner and more flavorful than domestic pork when the animal is young. Younger hogs under 100 lbs have the best meat quality. Older boars can have strong odor and require long, moist cooking. Always test for *Trichinella* and cook to 160°F internal temperature. Missouri has no closed season or bag limit for feral hogs — landowners and hunters are encouraged to eliminate them on sight. They root up native plant communities, destroy turkey and deer nesting sites, contaminate water sources, and cause millions in agricultural crop losses annually in Missouri.

Key Facts

- No closed season or bag limit in Missouri — actively encouraged to hunt and remove
- Cause estimated \$1.5 billion in agricultural damage annually in the US
- Straight tail (not curled) and large tusks on boars are key distinguishing features
- Rooting destroys native plant communities and creates erosion — invasive ecological disaster
- Always cook to 160°F internal temp — carry *Trichinella* and *Brucella*
- Sows can reproduce at 6 months and produce 2 litters/year of 4–8 piglets — eliminate quickly

Animal Tracking **Muskrat**



Muskrat (*Ondatra zibethicus*) are Missouri's most abundant semi-aquatic rodent — found in virtually every pond, marsh, and slow-moving waterway statewide. The tracks are 1–1.5 inches, 5-toed (5-toed hind, 4-toed front), with the distinctive tail drag creating a wavy center line between the track sets — the most reliable muskrat sign. Push-up lodges (mounds of cattail and aquatic vegetation 1–2 feet high in open water), runway channels through cattail marshes, and feeding platforms (flat mud mats with chewed vegetation) are characteristic signs.

■ Utility & Value

Muskrat are Missouri's most commercially important fur bearer by volume — hundreds of thousands are trapped annually. The dense, waterproof underfur is commercially valuable; prime pelts (December–February) fetch \$3–10 per pelt. The meat is edible — dark, mild, and similar to rabbit when prepared properly. Remove the scent glands before cooking. 'Marsh rabbit' is a traditional Missouri name for muskrat used on menus and in rural cookbooks. The push-up lodges provide winter habitat and feeding stations used by otters, mink, and diving birds.

Key Facts

- Tail drag line between track sets is most reliable muskrat track sign
- Push-up lodges: cattail-and-mud mounds in open water, 1–2 feet high
- Most commercially important fur bearer in Missouri by volume trapped
- Edible meat: 'marsh rabbit' in Missouri rural cooking tradition; remove scent glands first
- Feed platforms: flat mud mats with chewed cattail stems — active feeding sign
- Runway channels in cattail marshes ('eat-outs'): open water where muskrats have cleared vegetation

Animal Tracking **Raccoon**



The Raccoon (*Procyon lotor*) is one of Missouri's most adaptable and intelligent wild mammals, thriving in every habitat from dense Ozark forest to downtown Kansas City. The distinctive black facial mask and banded tail are unmistakable. Raccoons are omnivorous opportunists — crayfish, corn, eggs, fruit, garbage, pet food, and nearly anything else is on the menu. They use their dexterous front paws to manipulate food and open containers with remarkable precision. Nocturnal but commonly seen at dusk.

■ Utility & Value

Raccoon is a traditional Missouri wild game meat — particularly in the rural Ozark tradition. The dark, rich meat is best parboiled to tenderize, then roasted or braised with sweet potatoes and onions. Raccoon pelts were historically Missouri's most valuable fur and remain commercially traded. Raccoon hunting with hounds is a deep Missouri cultural tradition — particularly nighttime treeing with coonhounds. Raccoons can carry rabies and roundworm (*Baylisascaris procyonis*) in feces — never handle dead raccoons without gloves; their feces are a genuine health hazard particularly for children.

Key Facts

- Raccoon roundworm (*Baylisascaris*) in feces can cause serious neurological disease — never let children play where raccoons defecate
- Coonhound hunting (treeing) is a deeply rooted Missouri outdoor tradition
- Raccoon fat was traditionally rendered for cooking and waterproofing leather
- Dexterous hands can open door latches, cooler clasps, and childproof containers
- Missouri's raccoon pelt is commercially harvested — check current MDC trapping regulations
- Carry rabies, distemper, and parvovirus — do not approach disoriented daytime raccoons

Animal Tracking **Red Fox**



Red Fox tracks are small and neat, 1.5–2.5 inches long, oval with 4 toes and claws — very similar to Coyote but smaller and more compact. The defining feature is the 'bar' across the metacarpal pad — a rigid ridge of hair between the central pad and the toe pads that leaves a distinctive impression in soft snow or mud. Red Fox walks in an almost perfectly straight line (direct register gait) — a visual characteristic making a fox trail look like it was made by a single animal with two legs walking in a perfectly straight line through snow.

■ Utility & Value

Red Fox fur is among Missouri's most commercially valued — thick, luxurious winter pelts in prime condition (December–February) fetch \$15–50 per pelt. Foxes are targeted with live traps, snares, and calling. Understanding fox trails — particularly the characteristic straight-line trail across open fields and along fence rows — helps trappers and hunters identify regular travel routes. Red Fox are an important predator of cottontail rabbits, voles, and mice — their presence indicates a healthy small mammal population supporting the food chain.

Key Facts

- Track: 1.5–2.5 inch, oval, 4 toes with claws; 'bar' impression across heel pad in snow
- Perfectly straight trail line: most diagnostic tracking feature — nearly single-file dots in snow
- Direct register: hind foot lands in front foot print — extremely precise and efficient gait
- Scent marking: urine has a musky, skunky odor; deposited on prominent objects
- Fur prime December–February; track recognition key for trapper route planning
- Red Fox scat: dark, twisted, tapered at one end; filled with fur, feathers, or berry seeds

Animal Tracking **River Otter**



River Otter (*Lontra canadensis*) have fully recovered in Missouri after reintroduction programs in the 1980s–90s and are now widespread on Missouri's cleaner rivers and streams. The tracks are 2.5–3.5 inches wide, round to slightly oval, 5-toed with webbing, and claw marks. The diagnostic sign is the slide — where otters slide repeatedly on river banks (mud or snow), creating a 6–12 inch wide chute directly into the water. Spraint (scat) deposited on prominent rocks and logs has a strong fish odor and contains fish scales and crayfish fragments.

■ Utility & Value

River Otter have a regulated trapping season in Missouri (check current MDC regulations) and are among the most valuable Missouri fur bearers — prime pelts fetch \$40–100 or more per pelt. Their presence is a reliable indicator of excellent water quality — otters require clean water with abundant fish and crayfish populations. Finding otter slides along an Ozark stream means the watershed is healthy. Otter watching at dawn and dusk on clean Missouri streams is an outstanding wildlife viewing experience.

Key Facts

- Otter slides: 6–12 inch wide chutes into water on muddy or snowy banks — unmistakable sign
- Presence indicates excellent water quality — prime trout and smallmouth bass habitat too
- Spraint (scat) has strong fish odor; deposited on prominent rocks — scent-marking behavior
- Reintroduced to Missouri in 1982–1992 after historical extirpation
- Regulated trapping season — one of Missouri's most valuable fur bearers
- Often travel in family groups — multiple track sets and multiple slides at same location

Animal Tracking **Striped Skunk**



Striped Skunk (*Mephitis mephitis*) tracks are 1–1.5 inches wide with 5 toes and long front claws (digging tools) that leave prominent claw marks well in front of the toes — much longer claws than similarly-sized mammals. The front tracks show digging claws particularly well. The slow, plodding, pigeon-toed gait creates an unhurried track pattern. Cone-shaped holes in lawns and turf from grub-digging are the most commonly encountered skunk sign in suburban Missouri.

■ Utility & Value

Striped Skunk are commercially trapped for fur in Missouri — the soft, glossy black fur with white stripe is commercially processed with the scent glands removed. The musk (N-butyl mercaptan) can be smelled from a mile downwind and is extremely difficult to remove from clothing or pet fur. Neutralize with dilute hydrogen peroxide + baking soda solution — NOT tomato juice (this only masks, not neutralizes). Skunks are significant insect pest controllers — a family of skunks in a suburban yard may consume thousands of grubs per season, reducing lawn damage and Japanese Beetle populations.

Key Facts

- Long front claws leave exaggerated marks in tracks — key ID feature vs. similarly sized mammals
- Cone-shaped grub-digging holes in lawns = classic suburban skunk sign
- Neutralize skunk spray with H₂O₂ + baking soda + dish soap — NOT tomato juice
- Spray range: 10–15 feet with remarkable accuracy; only sprays as last resort after warning displays
- Slow-moving, confident gait — rarely flees from threats; the spray is the defense
- Significant grub-predator: reduces lawn grub damage and Japanese Beetle populations

Animal Tracking **Virginia Opossum**



The Virginia Opossum (*Didelphis virginiana*) is North America's only native marsupial and one of Missouri's most misunderstood mammals. Found statewide in forests, farmland, and suburban areas, it is the size of a large house cat with a pointed snout, naked prehensile tail, opposable rear thumbs, and 50 teeth — more than any other North American land mammal. Females carry young in a fur-lined pouch (marsupium) after an 13-day gestation — the shortest of any North American mammal.

■ Utility & Value

Opossum is edible and historically eaten throughout the rural South and Ozarks — 'possum and sweet potatoes' was a classic Depression-era Missouri dish. The meat is fatty and mild. They are extraordinary tick predators — a single opossum can kill and eat 5,000+ ticks per season through grooming, providing massive ecological disease control services. They are also immune to most snake venoms including Copperhead and Rattlesnake — peptides derived from opossum blood are being studied as potential antivenin treatments. 'Playing possum' (tonic immobility) is an involuntary catatonic state, not a choice.

Key Facts

- Consumes 5,000+ ticks per season through grooming — one of the best natural tick controllers
- Immune to most pit viper venoms — being studied for antivenin research
- Playing dead ('playing possum') is involuntary tonic immobility — they cannot control it
- North America's only marsupial — young born the size of a honeybee and crawl to the pouch
- 50 teeth — more than any other North American land mammal
- Rarely carry rabies — low body temperature makes survival of the virus difficult

Animal Tracking **White-tailed Deer**



White-tailed Deer tracks are heart-shaped split hooves (cloven hooves), 2–3.5 inches long, with the tips pointing in the direction of travel. Bucks' tracks tend to be wider and more rounded than does'; drag marks in snow indicate a heavy buck. The dewclaws (small rear hooves) show in soft mud or snow in a running deer. The walking gait shows paired tracks roughly 18–22 inches apart in a staggered pattern. Beds are oval depressions in leaves or grass, 3–4 feet long.

■ Utility & Value

Understanding deer sign — tracks, beds, rubs, scrapes, and trails — is fundamental to Missouri deer hunting and wildlife observation. Rubs (bark scraped from saplings by antlers) appear August–November and indicate buck territory. Scrapes (pawed ground with overhanging licking branch overhead) are primary rut communication sites, freshest in October–November. Large tracks crossing repeatedly indicate a primary travel corridor — funnel this into a stand location. The distance between paired hoof prints increases dramatically when deer are running — useful for tracking wounded deer.

Key Facts

- Track: heart-shaped split hoof, 2–3.5 inches; wider/rounder = buck
- Dewclaws visible in mud/snow when running or bounding
- Rubs on saplings: August–November; scrapes in leaves with overhanging branch: October rut peak
- Beds: oval 3–4 foot depressions in leaves; south-facing hillsides used in cold weather
- Scat: pellet-shaped, dark brown, 0.5–0.75 inch; clumped scat indicates softer food
- Trail cameras near scrapes in October reveal buck patterns — ideal stand placement tool

Animal Tracking **Wild Turkey**



Wild Turkey (*Meleagris gallopavo*) are a Missouri conservation success story — once nearly eliminated statewide by market hunting and habitat loss, they now number over 400,000 birds after Missouri's landmark restoration program beginning in the 1950s. The largest native North American game bird, adult toms reach 20–25 lbs with beard (modified chest feathers), sharp leg spurs, and the full fan-tail display used in spring courtship. The distinctive 'gobble' of spring toms carries over a mile in still air.

■ Utility & Value

Wild turkey is some of the finest wild game table fare available in Missouri — the dark, rich breast meat is far superior in flavor to commercial turkey. Missouri's spring gobbler season runs April–May (the peak of the rut when toms are vocal and actively responding to calls). Fall seasons for both sexes also exist. Wild turkey feathers were used by Native peoples for fletching arrows and ceremonial regalia. The drumstick meat can be slow-cooked for hours to tenderize. Young birds (jakes and jennies) have the most tender meat; old toms are best braised.

Key Facts

- Missouri turkey population: 400,000+ birds after landmark MDC restoration program
- Spring gobbler season: April–May — peak gobbling activity is early morning
- Tom turkey identifies by: beard, leg spurs (length indicates age), bright head colors in spring
- Wild turkey can fly 55 mph in short bursts and run 25 mph
- Outstanding eyesight: 270-degree field of view; can see color; nearly impossible to stalk
- Mast (acorn, hickory) is the primary fall/winter food — find mast, find turkeys

Trees

30 entries



American Elm (*Ulmus americana*) was once the signature tree of Missouri's towns and cities — massive, vase-shaped, arching over streets to form cathedral canopies. Dutch Elm Disease (*Ophiostoma ulmi*, introduced from Europe in the 1930s) killed virtually all large elms across Missouri and North America by the 1970s–80s. Scattered survivors and young trees still occur along Missouri's waterways and in forests. The elliptical, doubly-toothed, asymmetrical-based leaf is distinctive.

■ Utility & Value

Elm wood has interlocked grain making it nearly impossible to split along the grain — historically the preferred wood for wagon wheel hubs and chair seats that needed to resist splitting forces from all directions. The inner bark (bast fiber) was used by Indigenous peoples for rope and basket-making — it's fibrous, tough, and flexible. Elm seedlings and saplings still sprout vigorously along Missouri's waterways — Dutch Elm Disease-resistant cultivars have been developed and are being planted in street tree programs statewide.

Key Facts

- Dutch Elm Disease eliminated nearly all large American Elms by 1980 — a catastrophic ecological loss
- Interlocked wood grain: impossible to split radially — used for wheel hubs and chair seats
- Young elms and sprouts persist in Missouri waterways; mature trees now very rare
- Asymmetrical leaf base (one side higher than the other at the stem) is a key ID feature
- Disease-resistant cultivars ('Princeton', 'Valley Forge') being planted in urban programs
- Slippery Elm (*U. rubra*): inner bark is mucilaginous — traditional sore throat remedy



Bitternut Hickory (*Carya cordiformis*) is the most widespread hickory in Missouri, found in bottomland and moist upland woods statewide. It is uniquely identified by its bright sulfur-yellow winter buds — the most distinctive of any Missouri tree in winter. The 7–11 narrow leaflets and thin-husked, round nuts with bitter nutmeat complete the ID. It grows 50–80 feet in a broad range of soil conditions, from wet bottomlands to dry uplands.

■ Utility & Value

The nutmeat is intensely bitter — eaten only by wildlife (deer, squirrel, turkey) when other mast is unavailable. However, the wood is identical in quality to other hickories — extremely hard and tough, excellent for tool handles, furniture, and firewood. The sulfur-yellow buds make Bitternut the easiest Missouri tree to identify in winter. The nuts were occasionally pressed for lamp oil and used medicinally (bark tea for rheumatism) by Native peoples.

Key Facts

- Sulfur-yellow winter buds: most distinctive winter ID feature of any Missouri tree
- Most widespread hickory in Missouri — tolerates wet to dry soil conditions
- Nuts too bitter for human consumption but important secondary wildlife food
- Wood quality equivalent to other hickories — prized for handles and fuel
- 7–11 narrow leaflets vs. Shagbark's 5 large leaflets
- Bark tea used historically for rheumatism and skin conditions

Black Cherry



Black Cherry (*Prunus serotina*) is one of Missouri's most valuable native timber trees — reaching 60–80 feet in rich upland woods statewide. The distinctive bark on mature trees is dark gray, broken into thick, rough, irregular scales resembling burnt corn flakes. The narrow, lance-shaped leaves smell of almond extract (cyanogenic glycosides) when crushed. Clusters of white flowers in spring; dark purple-black cherries 1/3 inch ripen August–September.

■ Utility & Value

Black Cherry lumber is one of the most valuable hardwoods in Missouri — the rich, reddish-brown wood darkens beautifully with age and is prized for fine furniture, cabinetry, and millwork. The tart, mildly bitter-sweet fruit can be made into wine, brandy, jelly, and syrup — the basis for Cherry Bounce, a traditional Ozark liqueur (cherries, sugar, and whiskey). The bark was a traditional cough suppressant (wild cherry bark syrup, still sold in pharmacies today, is based on this tradition). Berries are eaten by at least 50 bird species.

Key Facts

- Burnt cornflake bark on mature trees: most distinctive Missouri bark after Sycamore and Hackberry
- Timber value: one of the top 3 most valuable Missouri hardwoods along with Black Walnut and White Oak
- Cherry Bounce: traditional Ozark liqueur — pack cherries in jar, add sugar and whiskey, wait 3 months
- Wild Cherry Bark syrup: sold in pharmacies today — cough suppressant based on Indigenous tradition
- Leaves, bark, and seeds contain cyanogenic glycosides — do not eat leaves or seeds
- 50+ bird species eat the fall cherries — major wildlife food tree

Black Oak



Black Oak (*Quercus velutina*) is a large upland oak, 60–80 feet, found on dry, acidic, sandy or rocky soils throughout Missouri's Ozarks and northern counties. The deeply lobed leaves have bristle-tipped lobes (Red Oak group) and a glossy, dark green upper surface. The inner bark is bright yellow-orange (yielding quercitrin dye — the historical 'quercitron') — this inner bark color is the definitive Black Oak ID feature. Acorns are small (0.5–0.75 inch), half-covered by a deep cup with loose, fringed scales.

■ Utility & Value

Black Oak acorns are high-tannin but consumed by wildlife when preferred mast fails. The bark's yellow inner layer was historically important as a dye source (quercitron), used to produce golden-yellow fabric dyes before synthetic dyes. The wood is used interchangeably with Red Oak for lumber, flooring, and furniture. Black Oak hybridizes extensively with other Red Oak group members, making precise identification challenging — look for the yellow-orange inner bark as the confirmatory feature.

Key Facts

- Inner bark is bright yellow-orange — most reliable ID feature; cut a small branch to check
- Historically important quercitron dye source — yielded golden-yellow fabric color
- Dry, acidic, sandy or rocky soils are preferred habitat — Ozark sandstone ridges
- Red Oak group: acorns take 2 seasons to mature; high tannin content
- Hybridizes with Shumard, Scarlet, and other red oaks — hybrid identification is complex
- Wood interchangeable with Red Oak for all commercial uses



Black Walnut (*Juglans nigra*) is one of Missouri's most valuable native trees, reaching 70–100+ feet in rich bottomland and upland soils statewide. The compound leaves (15–23 leaflets) emit a distinctive spicy-citrus odor when crushed. Fruits are round, 1.5–2.5 inch green husks that turn black when ripe in September–October. The husks stain everything they touch an indelible dark brown.

■ Utility & Value

The nutmeat is intensely flavored — bolder and earthier than commercial English walnut — and prized in baked goods, ice cream, and candy. To harvest: let husks blacken and soften (or drive over them), remove husks wearing gloves, wash shells, cure for 2–3 weeks, then crack with a hammer. The tree produces juglone, a chemical that inhibits many plants growing nearby. Black walnut lumber is among North America's most valuable hardwoods — prized for gunstocks, furniture, and veneer. Husk tea was used as a natural dye and antifungal treatment.

Key Facts

- Husks stain skin brown for weeks — wear gloves when processing
- Juglone toxin kills tomatoes, apples, and many garden plants within 50–80 feet of roots
- Black walnut lumber: a single mature tree can be worth thousands of dollars
- Cure nuts 2–3 weeks after husking; store in shell up to a year in cool dry place
- Husk extract is a traditional remedy for ringworm and other fungal infections
- Squirrels bury more nuts than they find — walnut trees depend on squirrels for dispersal



Blackjack Oak (*Quercus marilandica*) is a small to medium oak (15–40 feet) of Missouri's driest, poorest, rockiest soils — Ozark sandstone outcrops, chert ridges, and infertile uplands. The distinctive club-shaped leaf — broad and three-lobed at the tip, narrow at the base — is unmistakable. Often associated with Post Oak (*Q. stellata*) in 'Cross Timbers' habitats in southwest Missouri. Where Blackjack and Post Oak dominate, the site is reliably dry and infertile — useful ecological indicator.

■ Utility & Value

Blackjack Oak produces small, bitter acorns with high tannin content — less valuable as food but consumed by wildlife when preferred mast is unavailable. The wood is small, scrubby, and of little commercial value, but burns hot and long — prized for firewood and smoking meat in the southwest Missouri Ozarks. Its presence reliably indicates thin, dry, infertile soil conditions — useful for hunters assessing terrain and for gardeners who need to understand site limitations before planting.

Key Facts

- Club-shaped leaf: broad 3-lobed tip narrowing to a stalk — unmistakable leaf shape
- Indicator species for poor, dry, thin soils in the Ozarks
- Often co-occurs with Post Oak — 'Cross Timbers' habitat in southwest Missouri
- Small and scrubby compared to other oaks — rarely exceeds 40 feet on poor sites
- Dense, heavy wood burns hot — prized for firewood despite small tree size
- Post Oak (*Q. stellata*): cross-shaped leaf; also a dry-site indicator paired with Blackjack

Boxelder



Boxelder (*Acer negundo*) is Missouri's most abundant maple — a fast-growing, weedy tree of stream banks, disturbed ground, and urban areas statewide. Unlike other maples, it has compound leaves (3–5 leaflets) resembling poison ivy (though the opposite leaf arrangement instantly distinguishes it). The V-shaped paired samaras are the familiar 'helicopter seeds' of fall. Tolerates flooding, drought, and urban pollution better than most trees.

■ Utility & Value

Boxelder sap can be tapped in early spring for syrup — sugar content is lower than Sugar Maple but the sap flows earlier and more abundantly. The syrup has a distinctive flavor some describe as butterscotch or caramel. The wood is soft and of little commercial value, but Boxelder with red-stained heartwood (caused by a fungal infection) is prized by wood turners for its dramatic color — 'Boxelder burl' and red-stained wood command premium prices in the art wood market. Seeds are primary food for Evening Grosbeaks in winter.

Key Facts

- Only Missouri maple with compound leaves — 3–5 leaflets; opposite arrangement distinguishes from Poison Ivy
- Sap can be tapped for syrup — earlier flow than Sugar Maple; butterscotch-like flavor
- Red-stained heartwood (Boxelder burl): prized by woodturners for dramatic color; high market value
- Seeds (samaras) are primary winter food for Evening Grosbeaks — a 'grosbeak tree'
- Aggressive colonizer of disturbed sites and stream banks statewide
- Boxelder Bug (*Boisea trivittata*): black and red bug that swarms on/in houses in fall — harmless

Bur Oak



Bur Oak (*Quercus macrocarpa*) is Missouri's most drought-tolerant and fire-resistant oak and historically the dominant tree of Missouri's prairie/forest interface. It grows 50–80 feet with a massive, spreading crown and deeply furrowed, fire-resistant bark up to 2 inches thick. The large acorn (up to 1 inch) with a distinctive deep, fringed 'burr' cap covering more than half the nut gives the species its name. Found in western and central Missouri in more open, drier habitats than White Oak.

■ Utility & Value

Bur Oak produces the largest acorn of any Missouri oak — relatively low in tannins and palatable with light processing. A critically important food source for deer, turkey, and squirrel in prairie and savanna habitats. The bark's extreme thickness (an adaptation to prairie fires) historically made Bur Oak the tree that 'held the line' at prairie/forest edges, giving rise to the Midwest's historic oak openings and savannas. Many Missouri towns sit under massive, centuries-old Bur Oaks that predate European settlement. The wood is similar in quality to White Oak.

Key Facts

- Most fire-resistant Missouri tree — thick bark evolved for prairie fire survival
- Largest acorn of any Missouri oak — deep burr cap covers half the nut
- 'Witness trees' in old survey records — some Missouri Bur Oaks exceed 400 years old
- Historically defined the prairie/forest edge across Missouri's western counties
- Low-tannin acorns valuable for wildlife and human food use
- Drought-tolerant — thrives where other oaks fail in dry western Missouri soils



Chinkapin Oak (*Quercus muehlenbergii*) is a distinctive oak of Missouri's Ozark limestone bluffs, dry rocky slopes, and dolomite glades — wherever calcareous bedrock creates shallow, well-drained, alkaline soils. It reaches 40–70 feet with coarsely toothed leaves resembling those of Chinkapin (a chestnut relative) — oval with large, rounded teeth and no lobes. The acorns are small to medium with a thin-scaled cup and are particularly sweet and palatable. Bark is gray and flaky.

■ Utility & Value

Chinkapin Oak produces some of the sweetest, lowest-tannin acorns of any Missouri oak — they are often eaten with minimal processing and are highly preferred by deer, turkey, squirrel, and wood duck. In rocky, calcareous Ozark terrain where wildlife food can be sparse, Chinkapin Oak mast production is a critical resource. The wood is used similarly to White Oak — heavy, hard, and rot-resistant. The bark has been used medicinally for its astringent tannins.

Key Facts

- Oval toothed leaves (not lobed) — most distinctive oak leaf in Missouri
- Confined to limestone and dolomite outcrops in Missouri's Ozarks
- Sweetest acorns of any Missouri oak — high palatability for wildlife and humans
- Critical Ozark wildlife food in rocky terrain where other mast trees are scarce
- Often found with Eastern Red Cedar on exposed Ozark glades and bluffs
- Acorns ripen in one season (White Oak group) — available in September–October



Eastern Red Cedar (*Juniperus virginiana*) is Missouri's most abundant native conifer and one of the state's most ecologically dominant trees on dolomite glades, abandoned fields, and rocky Ozark slopes. It is not a true cedar (genus *Cedrus*) but a juniper. The aromatic, reddish-brown heartwood and scale-like foliage are distinctive. It has expanded dramatically across Missouri's grasslands since European settlement suppressed natural fire regimes — woody encroachment is a major conservation issue.

■ Utility & Value

The small, blue-gray 'berries' (actually modified cones) of Eastern Red Cedar are the primary flavoring in gin — they can be used to season game meats and make gin-like liquor infusions. The berries are also eaten by Cedar Waxwings (named for this food source) and over 50 bird species. The aromatic heartwood is the traditional lining for cedar chests and closets — the natural oils repel moths. The dense, ever-green foliage provides critical winter cover and nesting habitat for songbirds and small mammals. The wood is used for fence posts and decorative items.

Key Facts

- Berries are the primary gin flavoring ingredient — can infuse vodka with cedar berries for gin-like spirit
- Cedar Waxwing named for their dependence on this tree's berries in winter
- Aromatic heartwood repels moths — traditional cedar chest and closet lining
- Encroaching on Missouri's prairies and glades — controlled burns needed to protect open habitats
- Dense year-round foliage = critical winter bird roost and nesting cover
- Fence posts last decades — naturally rot-resistant heartwood used throughout rural Missouri

Eastern Redbud



Eastern Redbud (*Cercis canadensis*) is Missouri's most spectacular spring-flowering native tree, erupting in brilliant magenta-pink flowers in March–April before the leaves emerge. The flowers coat the bark directly (a pattern called cauliflory) on branches and even the main trunk. Heart-shaped leaves follow; flat seed pods (legume family) develop and persist through winter. Found statewide at forest edges, along streams, and in understory positions. Oklahoma's state tree (not Missouri's, which is the White Oak).

■ Utility & Value

Redbud flowers are edible — pleasantly sweet with a mild bean-like flavor. Add to salads, use as garnish, pickle in vinegar, or make into a simple syrup. Young seed pods in early summer can be eaten like snap peas — tender and sweet. The flowers are high in vitamin C and antioxidants. The wood is colorfully grained (reddish-brown with streaks) but small diameter limits its commercial use; it is prized for small turned objects and decorative woodworking. One of the most important early-spring nectar trees for native bees.

Key Facts

- Oklahoma's state tree — NOT Missouri's (Missouri's is the Flowering Dogwood)
- Flowers are edible: sweet, mild flavor — use raw in salads or pickle in vinegar
- Young spring seed pods edible like snap peas before they toughen
- Flowers emerge directly from bark (cauliflory) before leaves — unique Missouri sight
- One of the top spring nectar sources for native bees, particularly bumblebees
- Peak bloom: late March–mid April — typically 2 weeks before the dogwood bloom

Flowering Dogwood



Flowering Dogwood (*Cornus florida*) is Missouri's official state tree — a small, elegant understory tree (15–30 feet) beloved for its large white (or pink) floral bracts in April–May and brilliant scarlet fall foliage and bright red berry clusters in fall. The 'flowers' are actually white petal-like bracts surrounding a small cluster of true flowers. Found statewide in forest understory, particularly in the Ozarks. Dogwood Anthracnose (*Discula destructiva*) has reduced populations in some areas.

■ Utility & Value

Dogwood berries (drupes) are eaten by at least 36 bird species in Missouri — particularly thrushes, robins, bluebirds, and waxwings refueling for fall migration. The berries are technically edible but bitter and astringent to humans; historically used medicinally (bark tea as a quinine substitute for malaria during the Civil War). The wood is extremely hard and dense — used for shuttle weaving bobbins, golf club heads, and tool handles where hardness and smoothness are required. The bark was used as a red dye and teeth-cleaning chew stick.

Key Facts

- Missouri's official State Tree
- White bracts are not petals — the true flowers are the small yellow cluster in the center
- 36+ bird species eat the red fall berries — critical for migrating thrushes
- Bark tea used as quinine substitute for malaria fever during the Civil War
- One of the hardest, densest native woods in Missouri — used for specialty tool handles
- Dogwood Anthracnose fungus threatens populations — more common in humid, shaded sites

Honey Locust

Honey Locust (*Gleditsia triacanthos*) is a native Missouri tree of floodplains and disturbed areas, reaching 30–70 feet. The most dramatic feature is the massive, branched thorns that cover the trunk and branches — clusters of 3–12 inch thorns that historically impaled livestock and pioneers. Thornless cultivars are widely planted as landscape trees. The long, twisted seed pods (10–18 inches) contain sweet, edible pulp surrounding hard seeds.

■ Utility & Value

The seed pod pulp is sweet, edible, and nutritious — historically eaten by wildlife and humans. The thick pulp around the seeds can be eaten fresh or dissolved in water for a sweet drink. The hard seeds require roasting or boiling to soften. Deer, cattle, and hogs eat the pods extensively in fall. The wood is extremely hard, durable, and rot-resistant — used for fence posts, furniture, and railroad ties. The thorns of wild trees were used as pins and fasteners by Indigenous peoples. The thornless cultivar 'Moraine' is the most common street tree in Missouri cities.

Key Facts

- Trunk thorns: 3–12 inch branching clusters — historically used as natural fence barbed wire
- Seed pods: 10–18 inch; sweet pulp is edible and nutritious (seeds require cooking)
- Deer and livestock eat pods heavily in fall — excellent hunting site near pod-dropping trees
- Wood extremely rot-resistant — fence posts last decades; highly durable hardwood
- Thornless landscape cultivars ('Moraine', 'Skyline') are most common Missouri street trees
- Flowers are a major early-summer nectar source — excellent bee forage tree

Kentucky Coffeetree

Kentucky Coffeetree (*Gymnocladus dioica*) is one of Missouri's most distinctive native trees — massive bipinnately compound leaves (2–3 feet long, the largest leaf of any Missouri native tree), stout twigs with no terminal bud, and thick, gray, rough-ridged bark. Found in rich bottomlands and floodplains primarily in central and western Missouri. It is one of the latest trees to leaf out in spring and first to lose leaves in fall. Large, thick, leathery seed pods (6–10 inches) persist through winter.

■ Utility & Value

The seeds were roasted and used as a coffee substitute by early settlers — the species epithet and common name reflect this history. However, all parts of the raw plant contain cytisine (a toxic alkaloid) — the pods, seeds, and even the pulp are poisonous unprocessed. Roasting neutralizes the cytisine, making the seed safe for use as a coffee substitute. The wood is extremely durable — fence posts in the ground for 100+ years. It is increasingly valued as a landscape tree for its architectural, coarse-textured winter silhouette.

Key Facts

- Largest compound leaf of any Missouri native tree — bipinnate, 2–3 feet long
- Seeds TOXIC when raw (cytisine) — must be roasted to neutralize; used as coffee substitute historically
- One of the latest trees to leaf out in spring — mid-May in Missouri
- Thick, leathery pods persist through winter providing visual ID cue
- Wood extremely rot-resistant — posts last 100+ years; more durable than Black Locust
- *Gymnocladus* = 'naked branch' in Greek — refers to the long leafless season

Mockernut Hickory



Mockernut Hickory (*Carya tomentosa*) is Missouri's most common hickory species, found in upland woods throughout the state. The name 'mockernut' comes from the thick husk and shell that yield a relatively small nutmeat — 'mocking' the harvester's efforts. It reaches 50–80 feet with compound leaves (7–9 large leaflets) that are fragrant when crushed. The nuts ripen in October, with thick, tan husks that split into 4 sections. The bark is shaggy but less dramatically so than Shagbark Hickory.

■ Utility & Value

The nutmeat, though small relative to shell size, is sweet and flavorful — eaten fresh or baked into pies and cookies. Indigenous peoples used all hickory nuts extensively, particularly for hickory milk (pounding crushed shells and nuts in water to extract a rich, nutritious fat). The wood is extremely tough and resilient — hickory is the standard for tool handles (axes, hammers, hatchets) precisely because it absorbs shock without splitting. Hickory firewood burns the hottest and longest of any common Missouri hardwood.

Key Facts

- Most abundant Missouri hickory — found in virtually every upland hardwood forest
- Name 'mockernut': thick shell yields small meat — 'mocks' the harvester
- Hickory wood standard for tool handles — shock absorption prevents splitting
- Burns hotter and longer than any other common Missouri firewood
- Fragrant leaves when crushed — helps distinguish from similar species
- Nuts ripen October; thick husks split at 4 seams when fully mature

Osage Orange



Osage Orange (*Maclura pomifera*) is a distinctive native tree of Missouri's prairies and river bottoms, reaching 30–50 feet with a short, gnarled trunk, stout spines, and the iconic large (4–5 inch), bumpy, green, brain-like fruit (hedge apples). It was used as living fencerows across Missouri's prairies before barbed wire — rows of tightly planted Osage Orange ('hedge') created impenetrable cattle barriers. These 'Osage hedgerows' remain visible across Missouri's agricultural landscape.

■ Utility & Value

Osage Orange wood is the most rot-resistant and hardest native wood in North America — a fence post of Osage Orange can last 50–100+ years in the ground. It is also the finest bow wood of any North American tree — Indigenous peoples, particularly the Osage Nation (for whom it is named), prized it above all others for bows. The wood burns extremely hot. Hedge apples are folk-claimed as insect repellents — studies show the fruit extract repels insects, but whole fruits placed in homes have limited practical effect. The yellow-orange wood is beautiful for woodworking.

Key Facts

- Most rot-resistant native wood in North America — fence posts last 50–100+ years
- Finest traditional bow wood — Osage bow staves are prized by primitive archery enthusiasts
- Trunk color: bright orange-yellow when freshly cut — a striking woodworking wood
- Prairie hedgerows of Osage Orange replaced by barbed wire after 1874; many rows still standing
- Named for the Osage Nation of Missouri and Arkansas who prized it for bows
- Hedge apple extract does repel cockroaches — but placing whole fruit in homes is minimally effective

Pawpaw



The Pawpaw (*Asimina triloba*) is Missouri's largest native fruit and one of North America's best-kept edible secrets. It grows as an understory tree or large shrub, typically 15–30 feet tall, forming dense clonal colonies (patches) along stream banks, rich bottomland woods, and shaded ravines statewide. The large, drooping leaves — up to 12 inches long — give a tropical appearance strikingly out of place in the Midwest. Dark maroon, bell-shaped flowers appear in April before the leaves, and the fruit ripens August through October.

■ Utility & Value

Ripe pawpaws have custard-like, yellow-orange flesh tasting of banana, mango, and vanilla — rich in calories, vitamin C, magnesium, iron, and antioxidants. They can be eaten raw, blended into smoothies, baked into bread or ice cream, or fermented. Indigenous peoples including the Osage and Shawnee depended on them heavily. The bark and seeds contain acetogenins, compounds under study for anti-cancer properties. Fruit doesn't ship well, so wild-harvested pawpaws aren't commercially available — finding them in the wild is genuinely rare and valuable.

Key Facts

- Fruit ripens Aug–Oct; squeeze-test for softness like a ripe avocado
- Leaves and twigs emit a strong green-pepper odor when crushed
- Zebra swallowtail butterfly larvae feed exclusively on pawpaw leaves
- Clonal patches can persist for centuries in the same spot
- Seeds are toxic — do not eat; bark tea was used historically as an insecticide
- Missouri populations concentrated in Ozark stream valleys and Mississippi bottomlands

Pecan



Pecan (*Carya illinoensis*) is a massive native tree reaching 70–100+ feet, native to Missouri's southeastern river bottoms — particularly the Missouri and Mississippi River floodplains and the Bootheel. It is the largest member of the hickory family and produces the most commercially valuable nut in North American agriculture. Wild trees produce smaller nuts than cultivated varieties but are reliably abundant in native stands. The large compound leaves (9–17 leaflets) are among Missouri's longest at up to 20 inches.

■ Utility & Value

Wild Pecan nuts are thinner-shelled and smaller than commercial varieties but often more flavorful — buttery, sweet, and rich. Missouri's native Pecan stands in the Bootheel and Missouri River bottoms produce abundant crops in good mast years. Shell and eat fresh, make pecan pie, pralines, candied pecans, or press for oil. Pecan wood is excellent for smoking meats — milder and sweeter than hickory but similarly aromatic. The tree is a valuable timber species; plantation cultivation of native Pecan varieties is an emerging Missouri agroforestry enterprise.

Key Facts

- Native to Missouri's southeastern river bottoms — not just a Southern tree
- Largest hickory family tree in North America; can reach 100+ feet
- Wild nuts: smaller and thinner-shelled than commercial, but often richer flavor
- Pecan wood smoke: sweeter and milder than hickory — excellent for poultry and pork
- Missouri River bottoms between Jefferson City and St. Louis support native pecan groves
- Cultivated pecan orchards in Missouri's Bootheel are an emerging agricultural industry

Persimmon



American Persimmon is fully covered in the Plants section. As a tree: it reaches 20–60 feet with the signature alligator-hide bark — deeply furrowed into small, blocky squares that are unmistakable in any season. Found statewide in dry upland woods, old fields, and roadsides. Often spreads into pastures from fence rows, forming persimmon thickets that are significant deer and turkey habitat.

■ Utility & Value

See American Persimmon in the Wildflowers & Plants section. Tree utility: the wood is extremely hard and dense — in the same genus as ebony (Diospyros) and shares similar properties. Historically used for golf club heads, weaving shuttles, billiard cues, and tool handles. Persimmon thickets in pastures and old fields are excellent deer and turkey hunting habitat — particularly from September through January when fruit production continues.

Key Facts

- Bark: deeply furrowed into small blocky squares — most distinctive Missouri bark pattern after Sycamore
- Wood in same genus as ebony (Diospyros) — similarly dense and fine-grained
- Persimmon thickets = excellent deer and turkey hunting sites in fall
- Fruit persists into winter — one of the last wild fruits available before snow
- Dioecious: separate male and female trees; only female trees bear fruit
- Pioneer species: invades old fields and pastures rapidly from parent trees

Sassafras



Sassafras (*Sassafras albidum*) is one of Missouri's most recognizable and fragrant trees — the only Missouri tree with three different leaf shapes on the same individual: unlobed oval, mitten-shaped (1 lobe), and 3-lobed. All parts emit a distinctive spicy-sweet root beer fragrance when bruised. It grows 20–60 feet at forest edges and old fields throughout Missouri. The root bark is the original natural flavoring for root beer, and a cultural icon of Missouri's folk herbal tradition.

■ Utility & Value

Sassafras root bark tea was one of the most popular medicinal beverages in early American history — a spring tonic and blood purifier in Ozark folk medicine. The active compound, safrole, was removed from commercial root beer in 1960 when the FDA determined high doses caused liver tumors in rats (the quantity in a cup of tea is considered safe by most herbalists). Filé powder — ground dried Sassafras leaves — is a traditional Creole seasoning and thickener for gumbo. Young leaves can be added to soups. Twigs make aromatic toothbrushes. The orange fall color is exceptional.

Key Facts

- Only Missouri tree with 3 different leaf shapes (unlobed, 1-lobe mitten, 3-lobe) on same tree
- Root bark = original root beer flavoring; safrole removed from commercial use in 1960 by FDA
- Filé powder: dried ground leaves; traditional Creole gumbo thickener and seasoning
- Chewing a young twig makes a natural toothbrush with pleasant flavor
- Root bark tea: historic spring tonic — moderate occasional use generally considered safe
- Brilliant orange fall color — rival to Sugar Maple on a sunny October Ozark hillside



Shellbark Hickory (*Carya laciniosa*) is the largest-fruited hickory in Missouri — the rounded nut (1–2 inches) is the biggest of any native hickory and is enclosed in a thick husk that splits into 4–6 sections. Found in rich bottomland forests and moist lowlands, particularly along Missouri's major rivers. Distinguished from Shagbark Hickory by its 7 (vs. 5) leaflets, preference for wet bottomland sites, and even shaggier bark. Sometimes called 'Kingnut Hickory.'

■ Utility & Value

Shellbark produces Missouri's largest and most nutritious hickory nut — rich, sweet, and high in fat and protein. The thick shell requires a vise or heavy hammer to crack, but the large nutmeat rewards the effort. Prime food source for squirrels, deer, and wood ducks in Missouri's river bottoms. The wood is identical to Shagbark Hickory — extremely tough and resilient, used for tool handles, sporting goods, and premium firewood. The wet bottomland habitat is increasingly rare in Missouri due to drainage.

Key Facts

- Largest hickory nut in Missouri — 1–2 inch; prized as 'Kingnut'
- 7 leaflets (vs. Shagbark's 5) and bottomland habitat are the key distinctions
- Thick husk splits into 4–6 sections; thick shell requires heavy cracking
- Wet bottomland forest habitat — found along Missouri's major river floodplains
- Wood quality = Shagbark: toughest, most shock-resistant Missouri hardwood
- Disappearing habitat: Missouri bottomland forests heavily converted to agriculture



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Shortleaf Pine



Shortleaf Pine (*Pinus echinata*) is Missouri's only native pine and the characteristic tree of the Ozark Highlands' dry, cherty soils — particularly in the 'Missouri Pinewoods' of the St. Francois Mountains and adjacent counties. Reaching 60–90 feet with paired or triple needles 3–5 inches long, it forms open, savanna-like forests on dry ridges. The distinctive prickly cones (echinate = prickly) are 1.5–2.5 inches. Once covering millions of Ozark acres, it was heavily logged in the late 1800s–early 1900s.

■ Utility & Value

Shortleaf Pine timber was extensively harvested for railroad ties, mine timbers, and construction lumber in Missouri's Ozarks — this harvest, combined with fire suppression, reduced the pine coverage dramatically. Missouri's restoration efforts at Peck Ranch, Stegall Mountain, and other conservation areas are rebuilding pine communities using prescribed fire. The resin is edible and was chewed by Indigenous peoples; bark tea was used for respiratory ailments. The open, sunny pine forests support unique plant communities including rare Ozark wildflowers.

Key Facts

- Missouri's only native pine — confined to dry, cherty Ozark soils
- St. Francois Mountains and Reynolds County support the best remaining Missouri pinewoods
- Needles in bundles of 2 (sometimes 3); 3–5 inches long
- Requires fire to maintain open savanna structure — MDC uses prescribed burns for restoration
- Resin edible; chewed as gum and used for respiratory ailments traditionally
- Peck Ranch Conservation Area: one of best places to see restored Shortleaf Pine forest in Missouri

Shumard Oak



Northern Red Oak (*Quercus rubra*) and Shumard Oak (*Q. shumardii*) are the dominant upland oaks of Missouri's Ozarks and river bluffs. Both have pointed, bristle-tipped leaf lobes (the defining feature of the Red Oak group) and produce large acorns that take two full seasons to mature. Red Oak reaches 60–90 feet with a straight trunk and brilliant scarlet fall color. Shumard Oak is similar but larger-leaved and more common on bluff bases and bottomland edges. Both are highly valuable timber trees.

■ Utility & Value

Red Oak group acorns are higher in tannins than White Oak group and require longer leaching (cold-water soaking for 3–7 days, changing water twice daily) to become palatable. Once processed, the flour has an excellent reddish-brown color and rich, earthy flavor. The wood is ring-porous, strong, and red-toned — used for flooring, cabinets, and lumber. Red Oak lumber is slightly less weather-resistant than White Oak (open vessels in the wood) but is often preferred for interior millwork. Acorns are an important secondary mast food for wildlife when White Oak production is poor.

Key Facts

- Bristle-tipped lobes on leaves = Red Oak group (all produce higher-tannin acorns)
- Acorns take TWO seasons to mature — small acorns on twigs are year-one, full size are year-two
- High-tannin acorns require 3–7 days cold-water leaching before use as food
- Brilliant scarlet fall color — Red Oak is among Missouri's finest fall color trees
- Red Oak lumber is more porous than White Oak — less suitable for barrels and outdoor use
- Missouri's Ozark ridgetops support nearly pure Red Oak stands in some areas

Silver Maple



Silver Maple (*Acer saccharinum*) is the most common maple in Missouri, found throughout the state in bottomlands, floodplains, stream banks, and moist lowlands. It grows rapidly — 60–80 feet in 30–40 years — with deeply cut leaves silvery-white on the underside (the 'silver' of the name, visible when leaves flutter in the wind). One of the first trees to flower in Missouri — clusters of small red flowers appear in February before any other native tree blooms.

■ Utility & Value

Silver Maple sap can be tapped for syrup in late winter — it has lower sugar content than Sugar Maple (1–1.5% sugar) and requires more sap per gallon of syrup, but it flows more abundantly. The wood is soft and light — used for boxes, pallets, and pulp. Its rapid growth makes it popular as a landscape tree, but the weak wood breaks easily in ice storms and high winds. The samaras (winged seeds) falling in spring are eaten by squirrels and the February flowers are a critical early nectar source for native bees emerging from winter.

Key Facts

- First native Missouri tree to flower — February blooms before any other woody plant
- Silvery leaf undersides shimmer in wind — best ID character in summer
- Can be tapped for syrup: lower sugar content than Sugar Maple but abundant sap flow
- Fast-growing but brittle — branches break readily in ice storms; avoid planting near structures
- Most common maple in Missouri floodplains and stream banks
- Spring samaras eaten by squirrels; February flowers are early-season bee food

Slippery Elm



Slippery Elm (*Ulmus rubra*) is a medium-sized tree (40–70 feet) found in rich, moist upland and bottomland woods throughout Missouri. Similar to American Elm but with a rougher, sandpaper-like leaf surface (rough on both sides vs. American Elm's smoother upper surface). The inner bark (bast) is red-tinged and, when stripped and chewed, produces a slippery, mucilaginous texture — the source of its name and its primary medicinal use.

■ Utility & Value

The inner bark of Slippery Elm is one of North America's most important medicinal plants. The mucilage (primarily mucopolysaccharides) soothes irritated mucous membranes — throat lozenges containing Slippery Elm extract are widely sold and clinically effective for sore throats, coughs, and gastritis. Strip young bark, dry, and grind into powder to mix with water as a soothing gruel — historically a survival and convalescent food. The outer bark was used as emergency food (inner soft bark layer eaten raw or boiled) and for making rope, baskets, and mats.

Key Facts

- Inner bark produces thick mucilage — effective sore throat and GI soothing remedy
- Sold commercially as throat lozenges; rated 'likely effective' by Commission E (German herbal authority)
- Rough on BOTH leaf surfaces vs. American Elm rough only on top
- Inner bark strips can be eaten raw or dried and powdered as survival food
- Less affected by Dutch Elm Disease than American Elm — still relatively common
- Traditional Ozark remedy: inner bark tea for sore throat, ulcers, and diarrhea

Sugar Maple



Sugar Maple (*Acer saccharum*) is the iconic maple of Missouri's rich, moist upland woods and north-facing slopes — most abundant in the northeastern Ozarks and along Missouri's river bluffs. Missouri is near the southern edge of its primary range. The classic 5-lobed leaf with U-shaped sinuses and bright orange-red-yellow fall color is one of North America's finest autumn displays. It reaches 60–80 feet in Missouri's best sites.

■ Utility & Value

Sugar Maple is the primary source of maple syrup — it has the highest sugar content in its sap (2–3%) of any Missouri tree. Tap trees 8+ inches diameter in late winter (January–March) when temperatures freeze at night and thaw above 40°F during the day. Collect sap in buckets; it takes approximately 40 gallons of sap to produce 1 gallon of syrup. Missouri's Sugar Maple syrup tradition is smaller than New England's but well-established in the Ozarks. The hard, dense wood is one of North America's finest hardwoods — used for butcher blocks, dance floors, bowling alleys, and fine furniture.

Key Facts

- Tap in January–March: freeze nights + 40°F+ days = sap flow; 40 gallons sap = 1 gallon syrup
- Tree must be 8+ inches diameter before tapping; use 5/16 inch drill bit, 1.5 inch deep
- Missouri's finest fall color — brilliant orange, red, and yellow
- Wood: one of the hardest native American hardwoods — standard for butcher blocks and dance floors
- Missouri near southern range limit — found mainly in northeastern Ozarks and river bluffs
- Silver Maple (*A. saccharinum*) is a bottomland species; lower sugar content but also tappable

Sycamore



American Sycamore (*Platanus occidentalis*) is Missouri's largest native deciduous tree by trunk diameter — specimens along Missouri's major rivers can reach 10+ feet in circumference. The mottled, patchy bark exfoliating in cream, tan, and gray-green plates is the most distinctive bark pattern of any Missouri tree. Found along all major Missouri rivers and streams, often the largest tree visible in the floodplain. Ball-shaped seed heads persist through winter.

■ Utility & Value

Sycamore is valuable for its hollow trunk cavities — large old trees are critical nesting habitat for Wood Ducks, Eastern Screech-Owls, Great Horned Owls, and Raccoons. The wood is hard to work (interlocked grain) but used for butcher blocks and food containers (the grain doesn't impart flavors to food). The bark was used medicinally — inner bark tea for dysentery and as a cold remedy. The massive floodplain presence of Sycamore along Missouri's rivers makes it a critical visual landmark and navigation tree.

Key Facts

- Largest native deciduous tree in Missouri by trunk diameter — some exceed 10 feet around
- Mottled cream-and-gray exfoliating bark: most distinctive bark pattern in Missouri
- Hollow old trees are critical cavity nesting habitat for Wood Duck and large owls
- Found along virtually every Missouri river and stream of any size
- Wood's interlocked grain resists splitting — used for butcher blocks and food containers
- Ball-shaped seed clusters persist through winter — useful winter ID

White Ash



White Ash (*Fraxinus americana*) was once one of Missouri's most common and commercially important hardwoods — now facing functional extinction from the Emerald Ash Borer (*Agrilus planipennis*), an invasive beetle accidentally introduced from Asia. Once established, Emerald Ash Borer kills nearly 100% of untreated ash trees within 3–5 years. White Ash reaches 60–80 feet with compound leaves (5–9 leaflets), distinctive diamond-patterned bark, and paddle-shaped samaras. Green and Black Ash species are similarly affected.

■ Utility & Value

White Ash wood is one of the finest hardwoods in North America — tough, resilient, and shock-absorbing. It was the universal wood for baseball bats, hockey sticks, tennis racquets, tool handles, and oars. Most professional baseball bats are made from White Ash. Missouri had billions of board feet of ash timber; most is now dead or dying. Dead standing ash is useful for firewood (it burns well even green) and can be milled before it deteriorates. The Emerald Ash Borer represents one of the most destructive invasive species events in Missouri forestry history.

Key Facts

- Functionally extinct in many areas due to Emerald Ash Borer — treat valuable trees immediately
- Wood used for baseball bats, hockey sticks, and tool handles — finest shock-resistant hardwood
- Diamond-patterned bark on mature trees is the key ID feature
- Emerald Ash Borer: shiny green beetle, S-shaped larval galleries under bark = death sentence
- Dead ash should be salvaged for lumber or firewood promptly — deteriorates within 2–3 years
- Ash seeds (samaras) are an important fall food for songbirds, particularly Purple Finch

White Oak



White Oak (*Quercus alba*) is the dominant and most ecologically important tree in Missouri's Ozark Highlands, reaching 60–100 feet with a broad, spreading crown on mature specimens. The distinctive leaf has rounded lobes (no bristle tips, unlike red oaks), turning deep wine-red to burgundy in fall. The acorns are sweet with low tannin content and ripen in a single season — critically important as wildlife food. White Oak timber is Missouri's premier hardwood, commanding top prices for flooring, furniture, barrels, and millwork.

■ Utility & Value

White Oak acorns are the sweetest of any Missouri oak — low tannin content makes them palatable with minimal processing. Shell and eat raw, roast, or grind into flour. Soak in cold water 1–3 days (changing water) to leach remaining tannins. Acorn flour makes excellent pancakes, bread, and porridge — a staple food for Indigenous peoples across the continent. The wood is used for whiskey barrels (the tight grain and tyloses in the vessels make it nearly watertight), flooring, and fine furniture. A mature White Oak can produce 20,000+ acorns in a good mast year — supporting deer, turkey, squirrel, blue jays, and dozens of other species.

Key Facts

- Rounded leaf lobes (no bristle tips) — key distinction from Red Oak group
- Low-tannin acorns: sweetest Missouri oak, can be eaten with minimal leaching
- Acorns ripen in one season (Red Oak group takes two) — ready in September–October
- White Oak whiskey barrels: tight grain makes vessels nearly liquid-proof
- Missouri state record White Oak exceeds 100 feet in height and 300+ years in age
- Old-growth White Oak savannas once covered millions of Missouri acres — now rare

Danger Zone

10 entries

Danger Zone Chanterelle vs. Jack-o'-Lantern

The Golden Chanterelle (*Cantharellus cibarius*) resembles the Jack-o'-Lantern (*Omphalotus olearius*). Both are bright orange-yellow and funnel-shaped. Critical difference: chanterelles have forking, blunt-edged ridges (not true gills) that run partway down the stem. Jack-o'-Lanterns have true, knife-blade-thin sharp gills.

■ Utility & Value

Chanterelles are prized for fruity, apricot-like aroma — highly valuable. Jack-o'-Lanterns grow in clusters at tree bases; chanterelles grow singly on the forest floor. Jack-o'-Lanterns glow faintly blue-green in darkness — check them in a dark room.

Key Facts

- Chanterelle: blunt forked ridges; Jack-o'-Lantern: sharp true gills
- Jack-o'-Lanterns glow in the dark — diagnostic feature
- Jack-o'-Lantern flesh is orange throughout; chanterelle flesh is solid white inside

Danger Zone Identifying Missouri's Venomous Snakes

Missouri's 5 venomous snakes are all pit vipers with shared characteristics: triangular (arrow-shaped) head distinctly wider than the neck; elliptical (cat-like) pupils; heat-sensing pit organs in a depression between the eye and nostril; thick, heavy body relative to length; and a single row of scales under the tail (non-venomous snakes have a double row). The rattle of rattlesnakes and the cotton-white mouth display of the Cottonmouth are species-specific identifiers. Missouri's 4 harmless water snake species (*Nerodia*) are the most commonly misidentified.

■ Utility & Value

Learning to confidently distinguish Missouri's venomous species from harmless lookalikes is one of the most valuable safety skills for outdoor use of Missouri lands. The key: never rely on color pattern alone (many harmless snakes are patterned). Use the triangle-head + elliptical-pupil + pit combination. In practice: if you encounter a snake and aren't certain, give it 6+ feet of clearance and back away slowly. No snake in Missouri is capable of pursuing a human — all bites result from accidental contact or intentional handling.

Key Facts

- All MO venomous snakes are pit vipers: triangular head + cat-eye pupil + facial pit
- Single scale row under tail = venomous; double row = non-venomous (check only if snake is dead)
- Cottonmouth white mouth display: unique to this species — no harmless snake mimics this
- Rattle: unmistakable; even a partial rattle of 1–2 segments is audible up close
- Water Moccasin (Cottonmouth) restricted to southern Missouri; most 'water moccasins' in MO are harmless *Nerodia*
- Never handle any snake you cannot 100% positively identify — even dead snakes can envenomate reflexively

Danger Zone Plant Poisoning / Poison Control

Missouri's most dangerous plant poisonings involve confusion between edible and toxic species in the same habitats. The highest-risk scenarios: Wild Carrot/Queen Anne's Lace confused with Water Hemlock or Poison Hemlock; Wild Garlic confused with Death Camas; Wild Grape confused with Moonseed; Pawpaw fruit confused with other species. Children are at highest risk from attractive berries — Nightshade, Pokeweed, Moonseed, and Bittersweet all produce attractive fruits in Missouri backyards.

■ Utility & Value

Poison Control: 1-800-222-1222 — available 24/7/365 with toxicology expertise. Call for any suspected plant ingestion, even if no symptoms yet. Take or photograph the plant for identification. For skin contact (Poison Ivy, Wild Parsnip, Poison Hemlock): wash thoroughly with soap and water and cover skin from sunlight immediately. For serious plant ingestions with systemic symptoms (confusion, seizures, difficulty breathing), call 911 — do not wait for Poison Control. Activated charcoal may be recommended by Poison Control for recent ingestions.

Key Facts

- Poison Control: 1-800-222-1222 — call for ANY plant ingestion, don't wait for symptoms
- Highest-risk confusions: Wild Carrot vs. Water Hemlock; Wild Garlic vs. Death Camas
- Children at highest risk from berries: Nightshade, Pokeweed, Moonseed, Bittersweet
- Wild Parsnip sap + sunlight = severe burns: cover from UV immediately and wash
- Poison Ivy smoke inhalation: call 911 — can cause life-threatening airway inflammation
- Take a photo of the plant for identification — critical for treatment decisions

Danger Zone Safe Foraging Rules for Missouri

Missouri offers extraordinary foraging diversity across all seasons — but several universal safety principles must be applied before eating anything from the wild. The most dangerous assumption in foraging is confidence based on partial ID. Every year, experienced foragers are seriously injured or killed by species they believed they knew. The plant families with the highest Missouri risk: Apiaceae (carrot family — contains both edible and deadly members in similar habitats), Amanita mushroom genus (contains both edible and deadly species), and wild onion/garlic lookalikes (Death Camas).

■ Utility & Value

The Missouri Forager's Core Rules: 1) Positive ID on ALL features — not just 'it looks like.' 2) When in doubt, don't eat it — and never eat 'probably.' 3) Start with small test portions of new species — some individuals have reactions even to safe species. 4) Eat only one new species at a time — if a reaction occurs, you'll know which one caused it. 5) Consult multiple sources — one field guide is not sufficient for edible plant ID. 6) Know your lookalikes for every species you forage. 7) Know your Poison Control number: 1-800-222-1222. 8) Never forage in areas sprayed with pesticides/herbicides. 9) Never forage from roadsides, railway margins, or industrial sites. 10) Leave enough for wildlife and plant reproduction — never strip a population.

Key Facts

- Poison Control: 1-800-222-1222 — program into your phone before every foraging trip
- The deadliest foraging mistakes: carrot-family lookalikes (Water Hemlock), Death Camas, Amanita mushrooms
- Start with 1 tablespoon of any new species — individual reactions vary even to safe edibles
- Never forage from roadsides, railway margins, industrial sites, or recently sprayed areas
- ID every feature, not just appearance — smell, habitat, season, stem structure all matter
- Leave 2/3 of any population behind — sustainable harvest maintains future food sources

Danger Zone Tick Bite Prevention & Removal

Missouri hosts multiple tick species of medical significance: Lone Star Tick (most abundant), American Dog Tick (*Dermacentor variabilis*, vector of Rocky Mountain Spotted Fever), Black-legged (Deer) Tick (Lyme disease vector), and Gulf Coast Tick. Ticks are active March–November with peak activity April–June and August–September. They quest by climbing vegetation and reaching with front legs for passing hosts — they do not jump or fly. All life stages (larva, nymph, adult) require blood meals.

■ Utility & Value

Prevention: Permethrin-treat clothing (not skin) for multi-day protection — kills ticks on contact. Apply DEET or Picaridin to exposed skin. Tuck pants into socks. Check entire body within 2 hours of outdoor activity, paying special attention to scalp, behind knees, armpits, groin, and behind ears. Shower within 2 hours of coming inside — this significantly reduces tick attachment. Removal: use fine-tipped tweezers, grasp close to the skin surface, pull upward with steady pressure. Do not twist, burn, or apply petroleum — these can cause tick to release disease-carrying saliva into wound. Clean bite area with rubbing alcohol after removal. Submit tick to TickReport.com for identification and testing.

Key Facts

- Permethrin on clothing: kills ticks on contact; remains effective through multiple washes
- Check within 2 hours of outdoor activity: scalp, behind knees, armpits, groin are favorite attachment sites
- Remove with fine-tipped tweezers: grasp close to skin, pull steadily — no twisting or burning
- Lyme disease transmission requires 36–48 hours of attachment — prompt removal is prevention
- Lone Star Tick: can cause alpha-gal syndrome (red meat allergy) — a potentially lifelong consequence
- TickReport.com: mail tick for identification and Lyme disease testing — know what bit you

Danger Zone True Morel vs. False Morel

Missouri's most-prized edible fungus — the true morel (*Morchella* spp.) — has several dangerous look-alikes, chief among them the False Morel (*Gyromitra esculenta*). A true morel has a hollow cap and stem continuous from top to bottom; cut it lengthwise and you see a single uninterrupted air space. The cap is fully attached to the stem at the base with a pitted, honeycomb-like surface rather than wrinkled, brain-like folds. True morels fruit March–May in Missouri.

■ Utility & Value

False morels (*Gyromitra*) contain gyromitrin, which metabolizes to monomethylhydrazine — toxic even after cooking or drying. Never eat a 'morel' without slicing it top-to-bottom first.

Key Facts

- True morel: hollow top-to-bottom; false morel: cottony/chambered interior
- False morel cap: reddish-brown, wrinkled/brain-like — NOT a honeycomb
- True morel cap is fully fused to stem; false morel cap may hang free
- Gyromitrin toxin survives cooking — false morels are never safe to eat
- Poison Control: 1-800-222-1222

Danger Zone Wild Grape vs. Moonseed

Wild grape (*Vitis* spp.) vines produce edible fruit, but Canada Moonseed (*Menispermum canadense*) produces highly toxic dark berries that look similar. Key: grape leaves toothed/lobed; moonseed leaves shield-shaped with petiole near center of leaf underside. Grape bark shreds in strips; moonseed bark smooth.

■ Utility & Value

Wild grapes are edible fresh, as jelly, juice, wine, or dried. Moonseed berries contain dauricine alkaloids causing cardiac/neurological toxicity — potentially life-threatening.

Key Facts

- Moonseed seed is crescent-shaped; grape seed is oval/pear-shaped
- Grape tendril curls from nodes; moonseed has no true tendrils
- Moonseed leaf petiole attaches near leaf center (peltate)
- Cross-section a berry and examine seed shape if uncertain

Danger Zone Wild Onion vs. Death Camas

Wild onion (*Allium canadense*) and Death Camas (*Anticlea elegans*) both emerge in spring with grass-like leaves from a bulb. The definitive test: crush any part and smell. Wild onion always smells like onion — Death Camas has no onion smell whatsoever.

■ Utility & Value

Wild onion is edible — bulb, leaves, and flowers. Death Camas contains zygacine and steroidal alkaloids causing cardiovascular and nervous system failure. Never harvest without the smell test.

Key Facts

- Onion smell = safe; NO onion smell = do not eat
- Death Camas flowers: creamy white with 6 petals and a V-shaped gland
- Zygacine alkaloids are cardiotoxic — no antidote; treatment is supportive only

Danger Zone ■ Mushroom Poisoning

Missouri's most dangerous mushroom poisonings involve Amatoxin-containing species — Death Cap (*Amanita phalloides*), Destroying Angel (*A. bisporigera*), and *Galerina marginata*. These contain alpha-amanitin, which irreversibly inhibits RNA polymerase II, halting protein synthesis in liver and kidney cells. A single Death Cap cap contains enough toxin to kill an adult. The insidious pattern: symptoms are DELAYED 6–24 hours after eating. An apparent recovery phase (day 2–3) while liver destruction continues silently gives false reassurance before fatal organ failure on days 3–5.

■ Utility & Value

Mushroom poisoning protocol: If mushroom ingestion is suspected — even with no symptoms — call Poison Control immediately: 1-800-222-1222. Save or photograph the mushroom. Do not wait for symptoms. With Amatoxin poisoning, by the time jaundice, abdominal pain, and organ failure appear, massive irreversible damage has occurred. Liver transplant is the only option for severe cases. Supportive care, activated charcoal (if given immediately), and experimental treatments (silibinin) may help in early cases. The most effective treatment is preventing the mistake — know the Death Cap and Destroying Angel before you pick anything in the *Amanita* genus.

Key Facts

- Poison Control: 1-800-222-1222 — call immediately for ANY suspected mushroom poisoning
- Amatoxin symptoms DELAYED 6–24 hours — do not wait for symptoms to call
- False recovery phase (days 2–3) while liver destruction continues silently
- Death Cap + Destroying Angel + *Galerina marginata* = 90%+ of fatal US mushroom poisonings
- Save or photograph the mushroom — identification is critical for treatment decisions
- No reliable home test for Amatoxins — lab analysis required for confirmation

Danger Zone ■ Snakebite Protocol

Missouri has 5 venomous snake species: Copperhead (statewide), Cottonmouth (southern Missouri), Timber Rattlesnake (Ozarks), Western Pygmy Rattlesnake (Ozarks), and Eastern Massasauga (very rare, north Missouri wetlands). All are pit vipers with triangular heads, elliptical pupils, and heat-sensing facial pits. Most Missouri snakebites occur between April and October, and the majority involve Copperheads. Approximately 7,000–8,000 venomous snakebites occur in the US annually; fewer than 10 are fatal with prompt treatment.

■ Utility & Value

Immediate protocol: Stay calm (panic accelerates venom spread through lymphatic system). Remove rings, watches, and tight clothing from the bitten limb before swelling begins. Keep the bite below heart level if possible. Call 911 or get to an emergency room immediately — do not drive yourself. The only effective treatment for pit viper envenomation is CroFab antivenin, available at Missouri hospitals. Do NOT apply tourniquet, do NOT cut and suck the wound, do NOT apply ice (constricts blood flow and concentrates venom), and do NOT use electric shock. Photograph the snake if possible without approaching it — identification helps guide antivenin dosing, but do not delay transport.

Key Facts

- Call 911 or get to ER immediately — antivenin (CroFab) is the ONLY effective treatment
- Do NOT: tourniquet, cut and suck, apply ice, use electric shock — these cause additional harm
- Copperhead: most common Missouri snakebite; rarely fatal in healthy adults with treatment
- Cottonmouth and Rattlesnake bites: more severe; higher antivenin doses often required
- Poison Control: 1-800-222-1222 — call while in transit to ER
- 5 venomous species in MO: Copperhead, Cottonmouth, Timber Rattler, Pygmy Rattler, Massasauga